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A review of ocean tidal current energy technology: Advances, trends, and challenges

Special Collection: Wave-Structure Interaction

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ABSTRACT

Rapid advancements in renewable energy technologies have been primarily driven by the increasing global energy demand for energy and the growing concerns over environmental pollution. Among these technologies, tidal current energy has achieved significant attention due to its high energy density and stable energy output. Compared to other renewable energy sources, such as solar and wind power, tidal current energy is more stable and predictable, offering considerable potential for integration into future energy systems. However, despite its promise, several key technological challenges must be addressed to enable the widespread development of tidal current energy technology. To address these challenges, this paper presents a comprehensive review of recent advancements in ocean tidal current energy technology, with particular emphasis on the analysis of the main types of energy harvesting devices and support structures. Furthermore, it evaluates representative tidal current energy devices based on their operational principles, technical features, and applications. By systematically reviewing mainstream hydrodynamic analysis methods, this paper evaluates their applicability and limitations in assessing the performance of tidal current energy devices. This assessment improves the accuracy of performance predictions and offers theoretical support for the optimal design of the energy harvesting systems. In addition, the paper addresses key issues in the development of tidal current energy technology, including technological feasibility, cost management, and environmental impact. These discussions provide a foundation for identifying emerging trends and major technical challenges in the field. Finally, the development prospects of tidal current energy technology are explored from the perspectives of technological advancement, economic viability, and sustainability. It is concluded that achieving widespread adoption of tidal current energy depends on factors such as technological maturity, effective cost control, and the balanced integration of environmental and social benefits.

I. INTRODUCTION

Global energy demand—an essential driver of modern societal functions—is steadily increasing, a trend particularly pronounced in developing and emerging economies.[1] Concurrently, the rapid growth in energy consumption has led to serious environmental challenges, with carbon emissions and accelerating climate change being among the most pressing concerns.[2] In response, there is a growing global consensus on the urgent need to develop renewable energy sources as sustainable alternatives to conventional fossil fuels. Among these, tidal current energy stands out due to its favorable characteristics, including wide distribution, long-term sustainability, high predictability, and abundant resource availability—attributes that distinguish it from other forms of ocean energy in terms of development potential.[3]

Tidal current energy technology has progressively advanced from theoretical research to practical implementation. Several demonstration projects and commercial pilot programs have been deployed globally, contributing to a steady increase in installed capacity.[4] The most widely adopted tidal current energy converters are vertical-axis and horizontal-axis turbines. In addition, a range of innovative devices—including oscillating hydrofoils, ducted turbines, tidal kites, and Archimedes screws—are gradually emerging in the market.[5] These innovations are primarily aimed at improving efficiency, reliability, and long-term sustainability.

Hydrodynamic analysis is a critical tool for evaluating the performance of tidal current energy devices in engineering applications. Currently, the four primary hydrodynamic analysis methods are widely employed: the blade element momentum (BEM) method, the vortex method, the computational fluid dynamics (CFD) method, and the actuator line method (ALM). Each method exhibits unique characteristics and varying degrees of accuracy, complexity, and computational cost. A central challenge in tidal current energy research lies in selecting and optimizing appropriate computational models to enhance device performance while minimizing design and operational costs. Furthermore, the development of tidal current energy technologies is influenced by a combination of factors—including technological maturity, economic viability, and sustainability—in addition to hydrodynamic performance.

Building on this background, the present paper provides a comprehensive review of recent advancements in ocean tidal current energy technology, with a particular focus on the types of tidal current energy devices, their operational principles, the current status of development, and the application of hydrodynamic analysis methods. In addition, the paper examines emerging development trends and key challenges in the field, especially in advanced areas such as array deployment and intelligent operations strategies, through a critical analysis of existing technological limitations. The study also investigates the primary barriers to the large-scale expansion of tidal current energy and proposes potential solutions aimed at promoting its widespread adoption by providing theoretical insights and technical guidance.

II. CLASSIFICATION OF OCEAN TIDAL CURRENT ENERGY DEVICES

Tidal current energy devices generate electricity by harnessing the kinetic energy of water flow to drive turbines, functioning similarly to wind turbines. These devices can be classified into two categories based on their technological characteristics: energy harvesting devices and support structures.[6] Energy harvesting devices include horizontal-axis turbines, vertical-axis turbines, oscillating hydrofoils, and other emerging configurations. Support structures, on the other hand, are typically categorized as pile-supported, gravity-based, or floating system (Fig. 1). Ongoing technological advancements have led to significant improvements in both structural design and energy conversion efficiency, thereby accelerating the innovation, optimization, and deployment of next generation tidal current energy devices.

A. Classification of energy harvesting devices

1. Axial turbine

Axial tidal turbines can be classified into two types—horizontal-axis and vertical-axis—based on the orientation of the turbine axis relative to the direction of water flow. Horizontal-axis tidal turbines (HATTs), whose rotational axis is aligned parallel to the flow direction, are generally optimized for unidirectional flow conditions. In contrast, vertical-axis tidal turbines (VATTs) feature an axis oriented perpendicular to the incoming flow, allowing them to harness energy from bidirectional or omnidirectional currents (Fig. 2).

The operation and structural components of HATTs closely resemble those of horizontal-axis wind turbines.[7] In this system, the kinetic energy of water flowing parallel to the turbine axis is converted into mechanical energy, which is subsequently transformed into electrical power as the blades rotate. HATTs offer several advantages, including stable power output, strong resistance to flow field disturbances, and a low start-up flow velocity.[8] However, a significant technical barrier to their widespread adoption lies in the complexity of their support structures. These structures often present challenges in terms of installation and maintenance, thereby increasing both operational difficulty and maintenance costs.

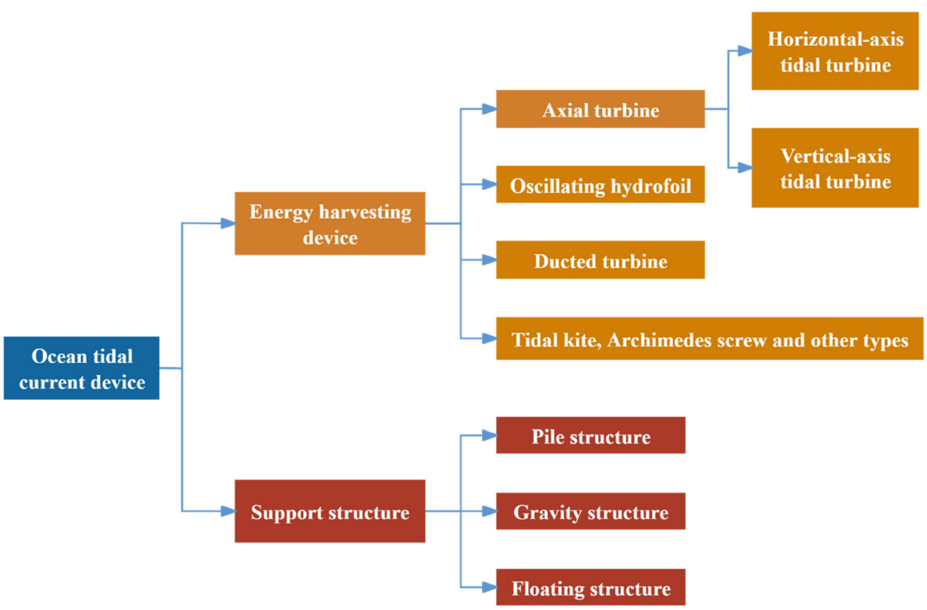


FIG. 1. Classification of ocean tidal current energy devices.

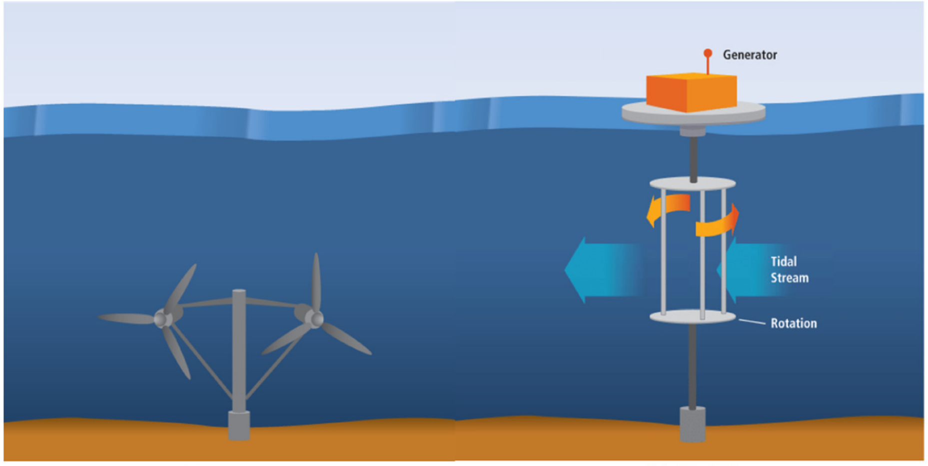


FIG. 2. Illustration of axial turbines.[10] [N. Hadžić, H. Kozmar, and M. Tomić, Brodogradnja **69**(2), 1–16 (2018); licensed under a Creative Commons Attribution (CC BY) license.]

VATTs are characterized by a rotational axis oriented perpendicular to the direction of water flow. Electricity is generated as the turbine rotates under the influence of lift and drag forces exerted by the water flow on the blades. A key advantage of VATTs over HATTs is their ability to operate effectively in bidirectional tidal currents without the need for a yaw mechanism. Additionally, the generator can be installed above the water surface, which reduces the complexity and cost associated with an underwater sealing system. However, VATTs typically demonstrate lower energy conversion efficiency due to the substantial variation in blade angles relative to the water flow during rotation. Furthermore, the turbine is subjected to complex and cyclically fluctuating hydrodynamic loads, which complicate structural design and impose stringent requirements on material fatigue strength and long-term durability.[9]

2. Oscillating hydrofoil

Oscillating hydrofoils generate electricity by converting the kinetic energy of tidal currents into oscillation motion. The fundamental operating principle involves hydrodynamic forces acting on a hydrofoil mounted on a fixed support. As water flows over the hydrofoil, variations in lift force induce a reciprocating oscillation about a pivot point.[11] This oscillatory motion is transmitted to a generator via a specialized drivetrain to produce electrical energy (Fig. 3).

Oscillating hydrofoils are particularly well-suited for shallow water environments, as their performance is not constrained by water depth. They offer several operational advantages, including a relatively simple mechanical structure, ease of maintenance, and stable energy output. However, the amplitude and frequency of the hydrofoil's motion are subject to physical limitations, which generally result in lower energy conversion efficiency compared to traditional turbines. Additionally, the reciprocating motion leads to increased mechanical wear over time, thereby raising maintenance costs. Environmental and ecological assessments are also essential, as the hydrodynamic disturbances caused by hydrofoil oscillations may impact local aquatic ecosystems.

3. Ducted turbine

A ducted tidal turbine incorporates an external duct that alters the water flow path to increase flow velocity, thereby enhancing energy conversion efficiency. Based on fluid dynamics principles, the device utilizes the geometric design of the duct to accelerate and direct the incoming water flow, optimizing its interaction with the turbine blades and increasing overall power output. The front curved section of duct acts similarly to a diffuser, modifying the surrounding flow field through the shaping of the duct walls. This design increases local energy density and flow velocity within the turbine's operational zone, resulting in improved performance (Fig. 4).

Ducted turbines can be classified as either unidirectional or bidirectional, depending on their functional characteristics. Unidirectional ducts, designed primarily for fixed flow directions, employ directional deflection and flow augmentation to enhance energy capture

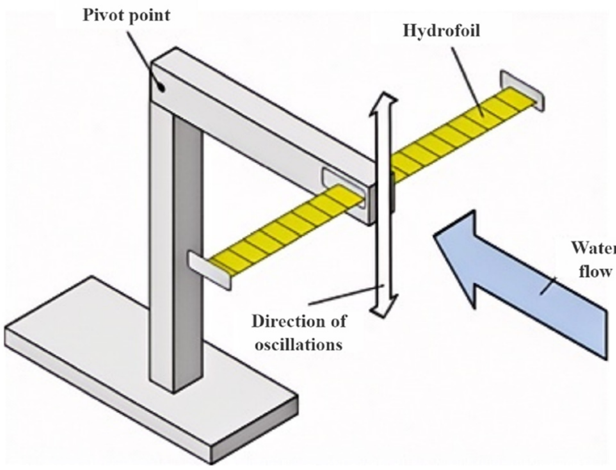


FIG. 3. Illustration of an oscillating hydrofoil.[12] [A. Roberts, B. Thomas, P. Sewell, Z. Khan, S. Balmain, and J. Gillman, J. Ocean Eng. Mar. Energy **2**, 227–245 (2016); licensed under a Creative Commons Attribution (CC BY) license.]

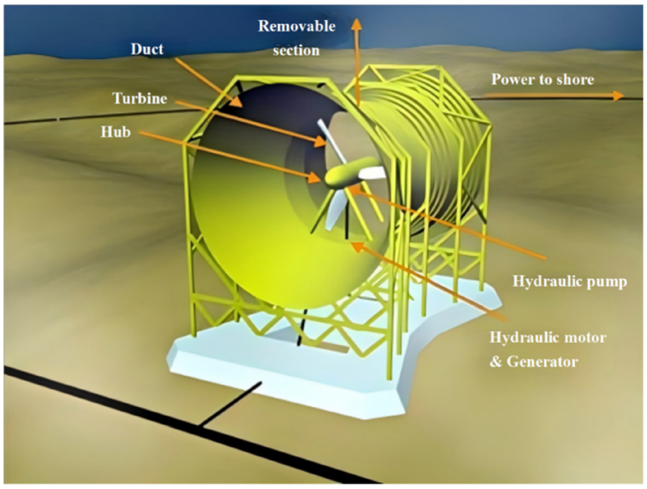


FIG. 4. Illustration of a ducted turbine.[13] (From Renewable Energy UK, 2025, https://www.reuk.co.uk/wordpress/tidal/lunar-energy-tidal-power/.)

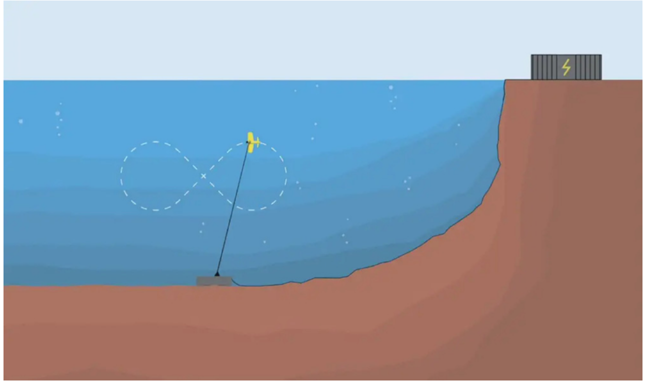


FIG. 5. Illustration of a tidal kite.[14] (From Offshore Engineer, 2025, https://www.oedigital.com/news/479125-faroe-islands-minesto-installs-foundation-for-tidal-kite-system.)

efficiency. Their relatively simple structural design makes them advantageous in stable flow environments. However, their performance significantly declines when the water flow direction changes, limiting their applicability in bidirectional tidal streams. In contrast, bidirectional ducts are capable of accommodating flow in both directions. These systems typically feature a symmetrical contraction–expansion structure, similar to a Venturi. Although the design is more complex, it allows for geometric optimization that maintains high energy conversion efficiency under varying flow conditions, making them well-suited for dynamic tidal environments.

The primary advantage of ducted turbines lies in their superior ability to converge water flow, thereby maximizing the pressure energy generated by the differential between upstream and downstream water levels and enhancing overall power generation efficiency. Additionally, by optimizing the design of the flow channel, ducted turbines effectively direct water flow toward the blade region. However, the manufacturing process is relatively complex, as the geometric configuration and structural integrity of the duct must be precisely calculated and optimized to ensure the device's reliability and stability in dynamic and harsh marine environments.

4. Other types

Tidal current energy devices include a diverse array of innovative designs beyond conventional turbines, such as tidal kites, Archimedes screws, and emerging systems like tidal fences. These alternatives offer new approaches to energy harvesting in various marine environments.

Tidal kites operate by following a periodic trajectory within the water column, utilizing lift generated by the water current acting on their wings. This lift-induced motion drives a generator, enabling the conversion of kinetic energy to electricity. The kite is maintained in stable "flight" within a designated depth layer by precisely controlling tether tension and rudder angle (Fig. 5). This technology demonstrates strong environmental adaptability, operational flexibility, and ease of maintenance, making it particularly suitable for deep-water applications. Furthermore, its unique attitude adjustment mechanism enables optimization of operational parameters based on the direction of tidal current, thereby enhancing energy capture efficiency.

The Archimedes screw harnesses tidal current energy through a spiral blade structure that rotates under the influence of flowing water (Fig. 6). Its operating principle is based on water entering the upper section of the device and descending along the spiral channel under the force of gravity. This rotational motion is then transmitted to a generator via the main shaft, facilitating mechanical-to-electrical energy conversion.[15] Characterized by large blades and a low rotational speed, the Archimedes screw allows debris, fish, and other aquatic organisms to pass through with minimal risk of harm. This feature offers a notable environmental advantage, making the device particularly suitable for ecologically sensitive marine and riverine environments.

A tidal fence generates energy by creating a water level differential through the strategic deployment of barriers that channel tidal flows through designated passages, where they drive power generation devices (Fig. 7). This technology is relatively mature and capable of

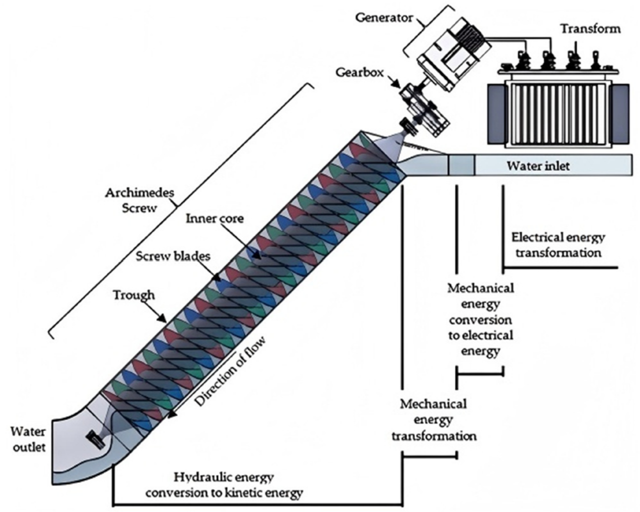


FIG. 6. Illustration of an Archimedes screw.[16] [Reproduced with permission from Edirisinghe **14**(11), 3307 (2021). Copyright 2021 Energies.]

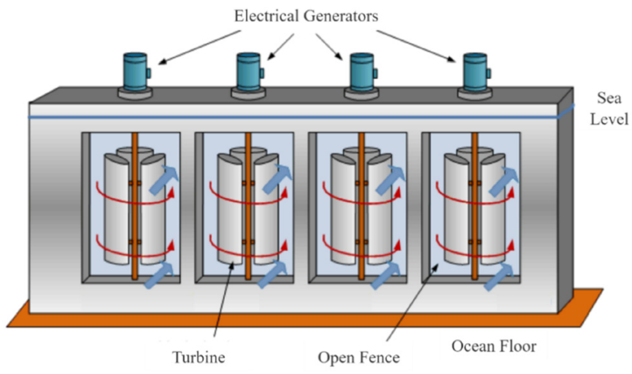


FIG. 7. Illustration of a tidal fence.[17] (From Alternative Energy Tutorials, 2025, https://www.alternative-energy-tutorials.com/tidal-energy/tidal-fence.html.)

providing stable power output, making it particularly suitable for regions with pronounced tidal fluctuations. However, tidal fences involve high construction costs and may significantly disrupt aquatic migration routes, leading to notable ecological impacts. Additionally, due to the inherently cyclical nature of tidal movements, the technology faces limitations in maintaining continuous power generation.

B. Classification of support structures

1. Pile structure

The pile structure functions as a foundational support system for tidal current energy devices, anchored directly to the seabed. Its fundamental design involves driving one or more steel pillars into the seabed to create a stable base, upon which tidal turbines and associated equipment are mounted.[18,19] This type of structure offers high operational stability and is minimally influenced by surface conditions such as wind and waves, owing to its direct fixation to the seabed. It also demonstrates strong adaptability to various seabed conditions, particularly in areas with solid geological formations. Furthermore, the compact and space-efficient design of pile structures enables the deployment of a higher density of generating units within a limited marine area, thereby enhancing power generation capacity per unit of sea surface area.

However, the installation and decommissioning of pile structures present significant challenges and require specialized offshore construction equipment, which substantially increases overall project costs. Additionally, the effectiveness of pile foundations is highly dependent on favorable subsea geological conditions. In regions with soft or unconsolidated seabed geology, the foundation may suffer from reduced bearing capacity or inadequate penetration depth, potentially compromising structural stability and elevating operational and maintenance risks.

2. Gravity structure

The gravity structure relies on a ballast-based foundation system, in which stability is achieved through the combined effect of the structure's self-weight and its interaction with the seabed. To resist slippage and overturning, a precast concrete gravity base is typically employed, enabling the tidal turbine and associated components to be securely positioned directly on the seafloor. Gravity structures are well suited for nearshore applications at water depths ranging from 20 to 80 m, provided the seabed possesses adequate load-bearing capacity.[20] These structures offer a stable and efficient platform for tidal current energy conversion in environments with favorable seabed conditions.

Compared to the pile structure, the installation of the gravity structure is relatively straightforward, as it does not require complex piling operations. This simplification results in shorter construction timelines and reduced installation costs. Additionally, gravity structures exhibit strong operational stability under conditions of large waves and high current velocities, due to their robust subsea anchoring and self-weight. However, their performance is sensitive to seabed topography. In areas with significant slope variations or irregular seabed profiles, uneven settlement may occur, potentially compromising the long-term structural stability. To mitigate these risks, comprehensive seabed assessments—such as multibeam bathymetry surveys and stratigraphic profiling—are essential prior to project deployment.

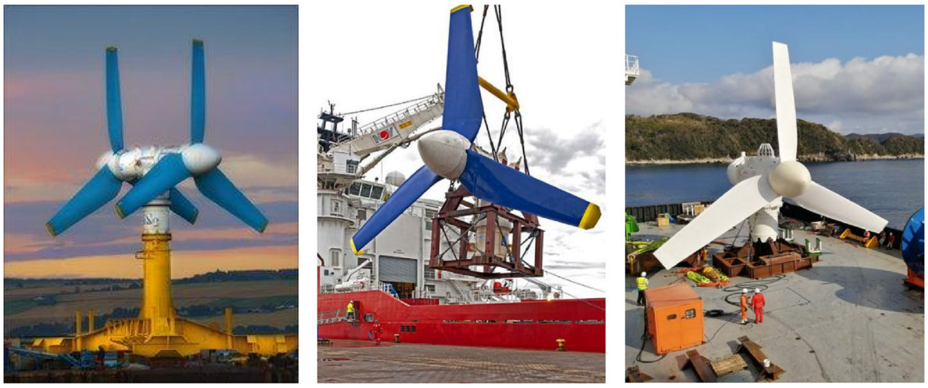
3. Floating structure

Floating structures are secured using a mooring system and are designed to integrate the tidal turbine and generator onto a floating platform. Precise positioning is achieved through a multi-point mooring configuration, allowing the structure to remain stable in dynamic marine environments. The design is particularly suitable for deep waters or regions or areas with complex seabed conditions. Floating platforms offer several advantages: they simplify installation and maintenance procedures, eliminate the need for complex underwater construction, and significantly reduce reliance on seabed geological conditions. Moreover, floating structures are capable of harnessing high-velocity water flows near the ocean surface, where flow conditions are typically more favorable. In contrast, flow velocity near the seabed is often reduced and unstable due to the influence of the undersea boundary layer effect.[21] By enabling optimal positioning within the water column, floating structures enhance energy capture efficiency and operational effectiveness.

However, under extreme sea conditions, floating structures are susceptible to six-degree-of-freedom motion. This can lead to structural fatigue and a reduction of power generation efficiency. To ensure reliable operation in such challenging maritime conditions, the deployment of an advanced power positioning system is required. Additionally, the integration of multi-degree-of-freedom mooring control and dynamic stabilization technologies is critical for enhancing stability and operational safety, and energy harvesting performance.[22–25]

III. DEVELOPMENT STATUS OF OCEAN TIDAL CURRENT ENERGY DEVICES

The rapid advancement of ocean energy technologies, coupled with ongoing innovation, has led to the emergence of a wide range of tidal current energy devices with practical engineering applications across the globe. In recent years, substantial progress has been achieved through demonstration projects and commercial pilot programs, reflecting the growing maturity of the sector. From conventional HATTs and VATTs to more novel approaches such as oscillating hydrofoils and ducted tidal turbines, these technologies have collectively driven innovation in both design and deployment strategies. Many of these devices have demonstrated clear technological advantages, including enhanced energy capture efficiency, innovative



(a) AK1000 turbine (b) AR1000 turbine (c) AR500 turbine

FIG. 8. SAE Renewables tidal device.[27] (From Power Technology, 2025, https://www.power-technology.com/.)

installation methods, and optimized structural configurations. At present, multiple representative tidal current energy devices have been successfully deployed in diverse maritime regions, where their technical feasibility and application potential have been validated through real-world operation. These deployments have generated valuable engineering insights and operational data, contributing significantly to the continued development and refinement of tidal current energy technology.

A. Horizontal axis tidal turbines

1. Pile structure

a. SAE tidal current energy device (Singapore). SAE Renewables, a Singapore-based company, introduced its AK series of HATTs in 2010[26] [Fig. 8(a)]. The series features a double-rotor configuration with fixed variable-pitch blades and is designed to operate at a rated flow velocity of 2.6 m/s. Each turbine stands 22.5 m tall and is available in 1 and 2 MW capacities, supported by a 1300-ton pile-based foundation. In May 2011, the 1 MW AR1000 turbine from this series was successfully connected to the grid at the European Marine Energy Centre (EMEC), and it entered commercial operation in July of the same year [Fig. 8(b)].

Further expanding its global deployment, SAE Renewables installed a 0.5-MW AR500 turbine in Japan's Naryu Strait in January 2021, demonstrating the company's capability growing capabilities and adaptability in the tidal current energy sector [Fig. 8(c)].

b. Sabella tidal current energy device (France). In 2015, the French company Sabella successfully deployed its full-scale D10-1000 demonstration unit in the Fromveur Passage.[28] This device features a horizontal-axis configuration with a six-bladed rotor, rated at 1 MW, and its fully submerged seabed-mounted installation (Fig. 9).

Since the commissioning of the demonstration project, Sabella has continuously refined the D10 turbine's technical performance, validating its operational reliability under extreme marine conditions. Supported by the PHARES program, the company is currently advancing the design and construction of the first commercial tidal current energy array in France. To promote international expansion and support the global development of tidal current energy technologies, Sabella has joined the TIGER experimental program in the Morbihan Gulf and established a strategic partnership with Nova Innovation in the United Kingdom.

c. Verdant tidal current energy device (the United States). The Roosevelt Island Tidal Energy (RITE) project, developed by U.S.-based Verdant Power, utilizes the TriFrame support structure, which accommodates three three-bladed HATTs.[30] This demonstration project set a national benchmark for ocean energy generation, having been successfully deployed in New York City's East River, where it generated a total of 210 MWh of electricity during its operational phase [Figs. 10(a) and 10(b)].

Looking ahead, Verdant Power plans to launch the Morlais tidal current energy project in 2026. This initiative is expected to become the largest tidal current energy generation site in Europe and the world's first large-scale tidal current energy development. In addition, to delivering clean, low-carbon electricity, the project aims to generate substantial economic and social benefits for the local community while making a meaningful contribution to global climate change mitigation efforts.

2. Gravity structure

a. Nova tidal current energy device (the United Kingdom). Between 2016 and 2017, Nova Innovation successfully deployed its first Nova

FIG. 9. Sabella D10 tidal device.[29] (From Ocean Energy Europe, 2025, https://www.oceanenergy-europe.eu/.)

(a) Verdant Power turbine (b) RITE project

FIG. 10. Verdant Power tidal device.[31] (From Verdant Power, 2025, https://verdantpower.com/.)

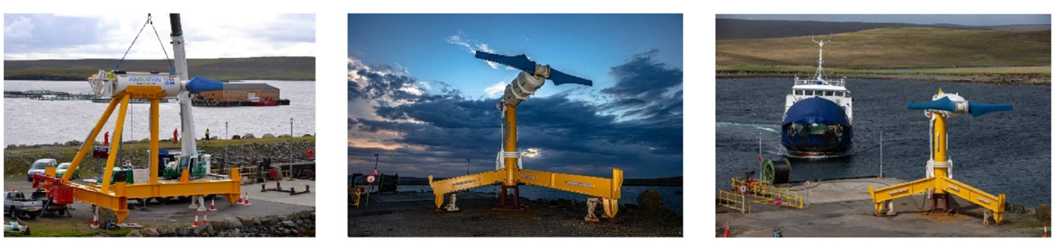
M100 turbines in the United Kingdom,[32] establishing the world's first grid-connected array of offshore tidal current energy devices [Fig. 11(a)]. By 2019, the array had accumulated over 17 000 h of operational runtime, demonstrating strong performance stability and reliability. In 2018, Nova Innovation, in collaboration with Tesla, integrated an energy storage unit into the system. This advancement marked the development of the world's first grid-connected tidal current energy power station with integrated energy storage, representing a significant milestone in the evolution of hybrid renewable energy systems.

To support the collection of full life-cycle operational data from the tidal current energy array, the government approved the removal of the three original M100 turbines and associated offshore infrastructure in 2023 as part of the EnFAIT (Enabling Future Arrays in Tidal) initiative. Following the decommissioning of these turbines in October 2023, the three M100-D turbines remained operational, continuing to supply power. The M100-D model features a two-bladed rotor design, securely supported by a steel gravity structure, with its drivetrain housed within a fully sealed, waterproof steel nacelle [Fig. 11(b)].

Building on this progress, the Seastar project aims to deploy 16 newly developed M500D turbines off the coast of the Orkney Islands, Scotland, forming a tidal current energy array with a total installed capacity of 4 MW [Fig. 11(c)]. This project represents a major milestone in the large-scale manufacturing and deployment of tidal current energy systems and highlights the growing maturity and engineering viability of tidal current energy technologies in commercial applications.

b. SAE tidal current energy device (Singapore). In 2013, the Singapore-based company SAE Renewables entered into a strategic partnership with Lockheed Martin to develop the 1.5 MW AR1500 turbine[26] [Fig. 12(a)]. The turbine was successfully commissioned as part of the MeyGen project and subsequently commercialized. In 2021, SAE Renewables announced the development of the 2 MW AR2000 turbine [Fig. 12(b)]. Featuring a gravity-based support structure and a horizontal-axis configuration, the AR2000 is designed to accommodate rotor diameters ranging from 20 to 24 m and operates at a rated flow velocity of 3.05 m/s. Engineering for broad applicability across diverse marine environments, the AR2000 represented the largest single-rotor tidal turbine in production at the time and incorporated an innovative pitch control system to enhance energy capture efficiency and operational flexibility.

c. Hammerfest tidal current energy device (Norway). In December 2011, the Norwegian company ANDRITZ HYDRO Hammerfest successfully installed a 1-MW HS1000 tidal turbine at the EMEC



(a) M100 turbine (b) M100D turbine (c) M500D turbine

FIG. 11. Nova Innovation tidal device.[33–35] [(a) From reNews, 2025, https://renews.biz/; (b) Nova Innovation, 2025, https://novainnovation.com/; and (c) Futuro Prossimo, 2025, https://en.futuroprossimo.it/.]

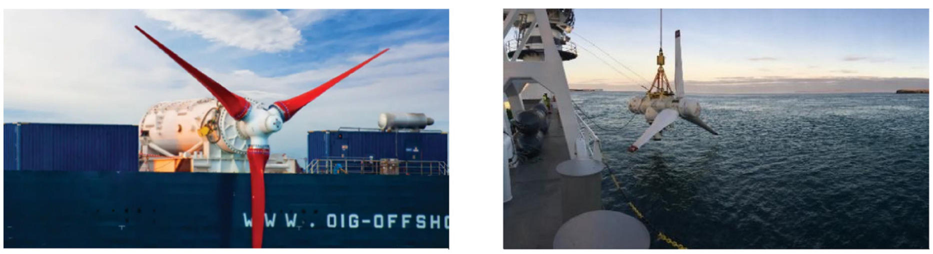
(a) AR1500 turbine (b) AR2000 turbine

FIG. 12. Renewables tidal device.[36,37] [(a) From Marine Technology News, 2025, https://www.marinetechnologynews.com/ and (b) SAE Renewables, 2025, https://saerenewables.com/.]

test site.[38] The turbine features a horizontal axis design with an 18 m-diameter rotor, optimized to achieve its rated power output at a flow velocity of 3 m/s [Fig. 13(a)]. The system incorporates an inclined blade structure, an integrated yaw mechanism, and a variable-speed gearbox, enabling efficient energy capture and operational flexibility. The HS1000 turbine is an advanced iteration of the earlier HS300 model, which was the world's first tidal turbine to be permanently connected to a public grid, initially deployed in Norway in 2004.

Building on the engineering insights and operational data obtained from the HS1000 turbine's EMEC testing, Hammerfest developed the next-generation 1.5 MW MK1 turbine [Fig. 13(b)]. Three MK1 turbines are currently in operation as part of the MeyGen project, located in the Inner Sound of the Pentland Firth, Scotland. As of February 2023, the MeyGen project has delivered over 50 GWh of electricity to the grid.

d. TGL tidal current energy device (the United Kingdom). At the EMEC test facility, the UK-based company Tidal Generation Limited (TGL) successfully deployed its 500 kW DeepGen III turbine and connected it to the grid in September 2010[40] [Fig. 14(a)]. By March 2012, the turbine had delivered over 200 MWh of electricity to the grid.



(a) HS1000 turbine (b) MK1 turbine

FIG. 13. Hammerfest tidal device.[37,39] [(a) From SAE Renewables, 2025, https://saerenewables.com/ and (b) Gurit, 2025, https://www.gurit.com/.]

(a) DeepGen III turbine (b) DeepGen IV turbine

FIG. 14. TGL tidal device.[27,41] [(a) From Power Technology, 2025, https://www.power-technology.com/ and (b) Wikipedia, 2025, https://en.m.wikipedia.org/wiki/Main_Page.]

Building on the performance of DeepGen III, TGL developed the second-generation 1 MW DeepGen IV turbine. This upgraded system features an innovative design that integrates a three-bladed variable-pitch rotor, a planetary gearbox, an induction generator, and an inverter system [Fig. 14(b)]. The turbine's buoyant nacelle design facilitates rapid installation and retrieval within a single tidal cycle, thereby significantly reducing operation and maintenance (O&M) costs. Additionally, the system incorporates an intelligent yaw mechanism that autonomously reorients the nacelle in real-time to align with varying tidal current directions, thereby maximizing energy capture efficiency and improving overall performance.

In 2013, TGL partnered with the Energy Technologies Institute to deploy the DeepGen IV turbine as part of the Reliable Data Acquisition Platform for Tidal program. The turbine is designed to be compatible with subsea wind technology and transmits power via a 6.6-kV wet connector, while utilizing the same tripod support structure as the earlier DeepGen III model. The DeepGen IV turbine achieved full power operation in July 2013 and surpassed 1 GWh of cumulative electricity generation by November 2014.

e. NENU tidal current energy device (China). The Northeast Normal University (NENU) in China has developed a horizontal-axis self-variable-pitch tidal current energy device designed to improve both operational efficiency and environmental adaptability (Fig. 15). A key innovation is its swing-tail tidal generator direction-adjustment mechanism, which enables the system to adapt effectively to bidirectional tidal current variations.[42] The turbine's horizontal-axis blade features an asymmetric wing structure, which enhances start-up performance and load-bearing capacity under low-flow conditions, while also reducing the overall structural weight. The device has an installed capacity of 20 kW, with a rated flow velocity of 1.5 m/s, a start-up flow velocity of 0.7 m/s, and an overall energy conversion efficiency of 26%.

FIG. 15. NENU tidal device.[43] (From Ocean Energy, 2025, http://www.hynyw.com/.)

It incorporates a 5-m blade diameter, is supported by an integrated tower support structure, and is deployed using an integrated drop-in installation method. In April 2013, the device successfully generated electricity in the waters near Zhaitang Island, Qingdao.

Under the State Oceanic Administration's major special funding project, titled "*Horizontal-Axis Self-Variable-Pitch Tidal Energy Engineering Prototype Design and Finalization*," NENU collaborated with Hangzhou JiangHe Hydro-Electrical Science & Technology Ltd. to significantly advance the technology. As part of this project, a 300-kW horizontal-axis tidal current energy generator set was successfully developed and achieved grid-connected operation in Zhejiang Province.

f. OUC tidal current energy device (China). The Ocean University of China (OUC) successfully developed the Haiyuan, a 100-kW-class horizontal-axis tidal current generator set[44] (Fig. 16). The device, with a rated operational flow velocity of 1.5 m/s and a total installed capacity of 100 kW, is deployed in a dual-unit configuration. To enable bidirectional tidal current self-adaptation and facilitate real-time operational monitoring and control, the system integrates variable-pitch control technology with a semi-direct-drive transmission system. For structural support, the device employs a tower-type fixed configuration with a gravity foundation, which enhances system stability by preventing multidimensional motion of the floating platform from affecting performance. This design also reduces the risk of fatigue damage caused by dynamic marine loads.

In August 2013, the system successfully completed a demonstration operation in the waters near Zhaitang Island, Qingdao. Power is transmitted to the central control center of the 500 kW Island Multi-Complementary Independent Power System via a 1-km underwater cable. Operational data indicate smooth and stable performance, with an overall energy conversion efficiency of 36.1%. The device can operate at startup flow velocities as low as 0.4–0.5 m/s when operating at the optimal pitch angle.

g. Hystrong tidal current energy device (China). Qingdao Hystrong Tower Ltd., based in China, has developed a 10-kW permanent magnet direct-drive tidal current energy device.[46] The device, featuring a gravity-based horizontal-axis structure, is designed for a rated

FIG. 16. OUC tidal device.[45] [Reproduced with permission from Y. Liu, **76**, 701–716 (2017). Copyright 2021 Renewable and Sustainable Energy Reviews.]

(a) First generation AQ turbine (b) Second generation AQ turbine

FIG. 17. Aquantis tidal device.[48] (From European Marine Energy Center, 2025, https://www.emec.org.uk/.)

flow velocity of 1.5 m/s, with an initial energy conversion efficiency exceeding 25%. Offshore validation tests conducted in 2014 confirmed the system's technical viability. The project emphasizes localized production of advanced tidal current energy components and integrated systems, combining cutting-edge technologies with established European engineering expertise. As a result, the project team has successfully achieved independent manufacturing of key components and complete systems, with reported energy conversion efficiency surpassing 30%. The device employs permanent magnet direct-drive technology, a system commonly used in large-scale wind turbines. Compared to conventional gearbox-based variable-speed power generation systems, this approach offers significant advantages in both efficiency and reliability. Operational data indicate that the system's power generation efficiency can be improved by 5%–10%, while also reducing internal energy consumption—demonstrating clear techno-economic benefits for future deployment and commercialization.

3. Floating structure

a. Aquantis tidal current energy device (the United States). The AQ series of tidal current energy devices, developed by the U.S.-based company Aquantis, marks a significant breakthrough in marine renewable energy technology.[47] The device features a unique floating structure anchored to the seabed via a mooring system, which ensures structural stability even under extreme marine conditions [Figs. 17(a) and 17(b)]. An upstream variable-pitch rotor, located below the water surface, drives the energy generation system, while all control and monitoring systems are positioned above the surface to facilitate operational safety and ease of maintenance.

Aquantis has developed proprietary turbine technology designed to deliver competitive lifetime energy costs, characterized by (1) a high power-to-weight ratio, (2) low deployment costs, and (3) simplified operation and maintenance. Building on this foundation, the company introduced the Tidal Power Tug, a second-generation floating tidal current energy converter. The system features a flexible spar-buoy platform that supports a two-bladed variable-pitch rotor with a 10 m diameter and an integrated 160 kW drivetrain. To ensure stable power generation across a range of sea conditions, the device is also equipped with a parallel-flow substructure, vertical yaw floats, and an advanced real-time analysis system.

b. Alstom tidal current energy device (the United Kingdom). In 2014, Alstom (formerly TGL) developed the Oceade turbine.[40] The system features an 18-m-diameter rotor, a rated power output of 1.4 MW, and an advanced three-bladed variable-pitch control system (Fig. 18). Its modular design incorporates plug-and-play functional units that allow for rapid assembly and maintenance via an access hatch located at the rear of the nacelle, thereby improving overall O&M efficiency.

The Oceade turbine utilizes a floating support structure, enabling easy towing and transport at sea, which significantly reduces installation and maintenance costs. An intelligent yaw system continuously monitors and adjusts the turbine's orientation in real time, ensuring

FIG. 18. Alstom Oceade tidal device.[49] (From Windpower Engineering & Development, 2025, https://www.windpowerengineering.com/.)

(a) SR250 turbine (b) SR2000 turbine (c) O2 turbine

FIG. 19. Orbital tidal device.[48,52,53] [(a) From European Marine Energy Center, 2025, https://www.emec.org.uk/; (b) The Guardian, 2025, https://www.theguardian.com/us; and (c) Orbital Marine Power, 2025, https://www.orbitalmarine.com/.]

that the rotor maintains an optimal angle of attack to maximize energy capture efficiency. The project was initiated in September 2013 under the French government's "*Invest in the Future*" program, following a national call for expressions of interest. It is planned for deployment at a tidal current energy site, where four Oceade turbines and an associated Alstom subsea electrical hub will be installed. The project's total installed capacity of 5.6 MW is expected to supply electricity to approximately 15 000 households. Offshore construction was initially scheduled to begin in 2017, with a designed operational lifespan of 20 years.

c. Orbital tidal current energy device (Scotland). Orbital Marine Power, a Scotland-based company, was the first in the world to achieve grid connection of a floating tidal current energy device.[50] Its first turbine, the SR250, featured a 250 kW and features a twin-rotor counter-rotating design [Fig. 19(a)]. The turbine includes 8 m rotor diameter, a 34 m hull length, and a gross weight of 100 tons, secured by a four-point suspended chain-line mooring system.[51] Constructed in Belfast in 2010, the SR250 was launched and tested in 2011, successfully connected to the Orkney electricity grid in 2012, marking a significant milestone in tidal current energy generation.

In 2016, Orbital Marine Power introduced the SR2000, a 2-MW turbine, which was the world's most powerful tidal current energy generator at the time [Fig. 19(b)]. The SR2000 featured two 16-m-diameter rotors, a 63-m hull length, and a total weight of 550 tons. It was deployed in 2016 and began supplying power to the grid in October of the same year. By the conclusion of its testing phase in 2018, the turbine had generated more than 3 GWh of electricity.

Building on the success of the SR2000, Orbital developed the O_2 turbine, featuring two 20-m-diameter rotors, a 72-m hull length, and a weight of 680 tons [Fig. 19(c)]. The O_2 turbine is equipped with 360° omnidirectional rotating blades, enabling efficient energy capture from bidirectional tidal flows. Successfully connected to the grid in 2021, the floating platform is anchored by a four-point mooring system, with power transmitted to the onshore grid via subsea cables. Orbital Marine Power is also working in collaboration with Lloyd's Register of Shipping to achieve IEC certification for the O_2 turbine, aiming to further advance the commercialization and technical standardization of tidal current energy technology. The O_2 turbine represents the first of a new generation of O_2-X devices, which are expected to achieve an installed capacity of 2.4 MW, contributing to the scalable deployment of renewable tidal current energy.

d. Magallanes tidal current energy device (Spain). Since 2007, the Spanish company Magallanes Renovables S.L. has been actively developing floating platforms and associated systems to harness tidal current energy and convert it into electricity.[54] One of its key innovations is the ATIR floating power generation platform, which features a novel design incorporating an integrated dual-rotor counter-rotating system, with a rated capacity of 1.5 MW (Fig. 20). The platform was successfully deployed at EMEC Fall of Warness test site in February 2019 and connected to the grid in March via submarine cables and onshore substations.

The ATIR platform has a total length of 45 m and a modular design, composed of three primary functional modules: the upper module, the vertical module (mast), and the lower module (nacelle). The upper module serves as the control center of the platform, housing critical components such as pumping units, transformers, converters, switchgear, and power distribution system, while also providing access for maintenance personnel. The vertical module acts as a structural support element, featuring a hollow design that facilitates the routing of communication and low-voltage cables between the upper and

FIG. 20. Magallanes ATIR tidal device.[55] (From Marine Energy Wales, 2025, https://www.marineenergywales.co.uk/.)

lower modules. The lower module contains the mechanical drive system, including the main shaft, ball bearings, gearbox, and generator. It is externally coupled to a three-bladed rotor, with each blade measuring 8.5 m in length.

e. SME tidal current energy device (the United Kingdom). The first PLAT-O system, developed by the UK-based company Sustainable Marine Energy (SME), was deployed at EMEC in 2016, following successful offshore trials in the Solent in 2015[56] [Fig. 21(a)]. The system features a submerged floating platform that dynamically adjusts its position in response to tidal current strength and incorporates 16 SIT turbines. Each turbine is equipped with a 4-m-diameter downstream rotor that includes automatic flow regulation to optimize performance. The PLAT-O system integrates several innovative technologies, including hydrodynamically optimized passive-adaptive composite blades, which enable efficient energy capture without the need for an active pitch mechanism. The drivetrain consists of a two-stage planetary gearbox, coupled with an induction generator and standard mechanical braking systems.

Building on this foundation, SME developed the next-generation floating platform, PLAT-I, which is equipped with four SIT250 tidal current energy turbines, delivering a total power output of 280 kW [Fig. 21(b)]. The platform successfully completed demonstration operations in Canada, becoming the first floating tidal current energy device to supply electricity to the Nova Scotia power grid. Currently, SME is collaborating with EMEC on the UMACK program, which focuses on developing a universal base mooring solution aimed at reducing installation and maintenance costs while enhancing the operational flexibility of floating tidal current energy platforms.

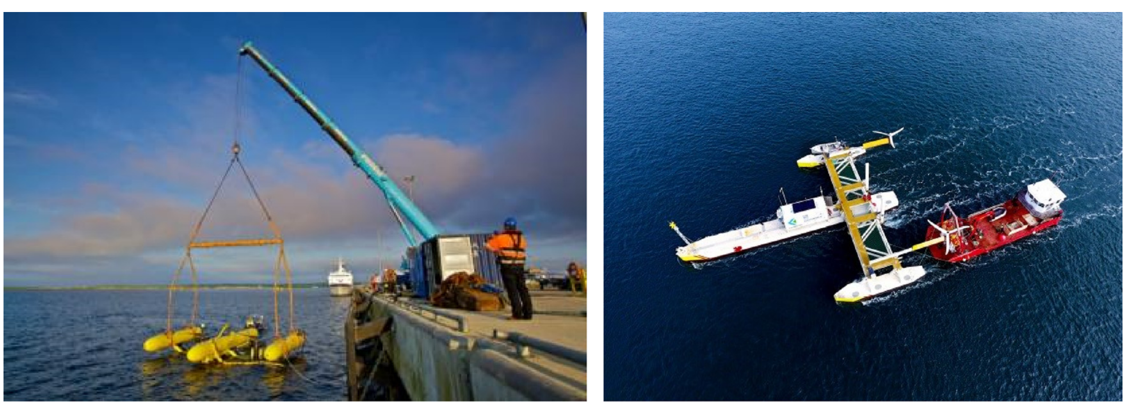
f. Oceanflow tidal current energy device (the United Kingdom). The development of the Evopod series—comprising twin and multi-turbine configurations—is being led by the UK-based company Oceanflow Energy.[58] As part of this initiative, the research team conducted scale model tests of a 2.4-MW twin-turbine solution in a flume tank at Newcastle University, and further evaluated its dynamic performance using the OrcaFlex mooring simulation software [Figs. 22(a)–22(c)].

In a parallel project, Oceanflow Energy is also developing Starfloat, a floating offshore wind platform that combines semi-submersible platform technology with a low-cost steel framework. The platform features a compact configuration that integrates ballast wing girders with a Multifloat semi-submersible buoyancy ring, enabling deployment in deeper waters than conventional single-wing girder designs. This approach aims to significantly reduce the levelized cost of energy (LCOE) while expanding the range of applicable offshore environments.

In August 2014, Oceanflow Energy successfully deployed the E35 tidal current energy device at Sanda Sound [Fig. 22(d)]. The system includes a 4.5-m-diameter turbine with a total weight of 12.5 tons, secured using a pre-laid mooring system. After nearly 1 year of offshore survival testing, the device demonstrated outstanding performance, exceeding expectations in terms of motion characteristics, heading control, and overall survivability of the semi-submersible platform. The Oceanflow Energy is currently working on grid connection procedures and plans to initiate power transmission upon completion of the submarine cable connection.

g. Tocardo tidal current energy device (Netherlands). The Tocardo T-1 turbine, developed by Tocardo International BV, a Netherlands-based company, represents a significant advancement in clean energy conversion technology[60] [Fig. 23(a)]. With a power output ranging from 50 to 100 kW, the turbine integrates Dutch hydraulic engineering expertise with renewable energy generation to deliver predictable and cost-effective energy performance. Specifically designed for shallow-water applications, the T-1 turbine features a lifespan of up to 20 years and requires minimal maintenance. In 2008, a demonstration unit was successfully deployed at the Afsluitdijk and connected to the Dutch national power grid.

In 2015, Tocardo introduced the T-2 series, increasing the power output per unit to 100–250 kW and significantly enhancing its suitability for commercial and array-scale deployments [Fig. 23(b)]. That same year, the company installed five T-2 turbines at the Eastern Scheldt Storm Barrier, achieving a peak power output of 1.2 MW at a flow velocity of 4.0 m/s. Located in one of the world's most energetic tidal current regions—where maximum flow reaches up to 6.0 m/s—this demonstration project provided an ideal testbed for evaluating turbine performance under extreme marine conditions, including corrosion, biofouling, and abrasion.



(a) PLAT-O platform (b) PLAT-I platform

FIG. 21. SME tidal device.[48,57] [(a) From European Marine Energy Center, 2025, https://www.emec.org.uk/ and (b) Baird Maritime, 2025, https://www.bairdmaritime.com/.]

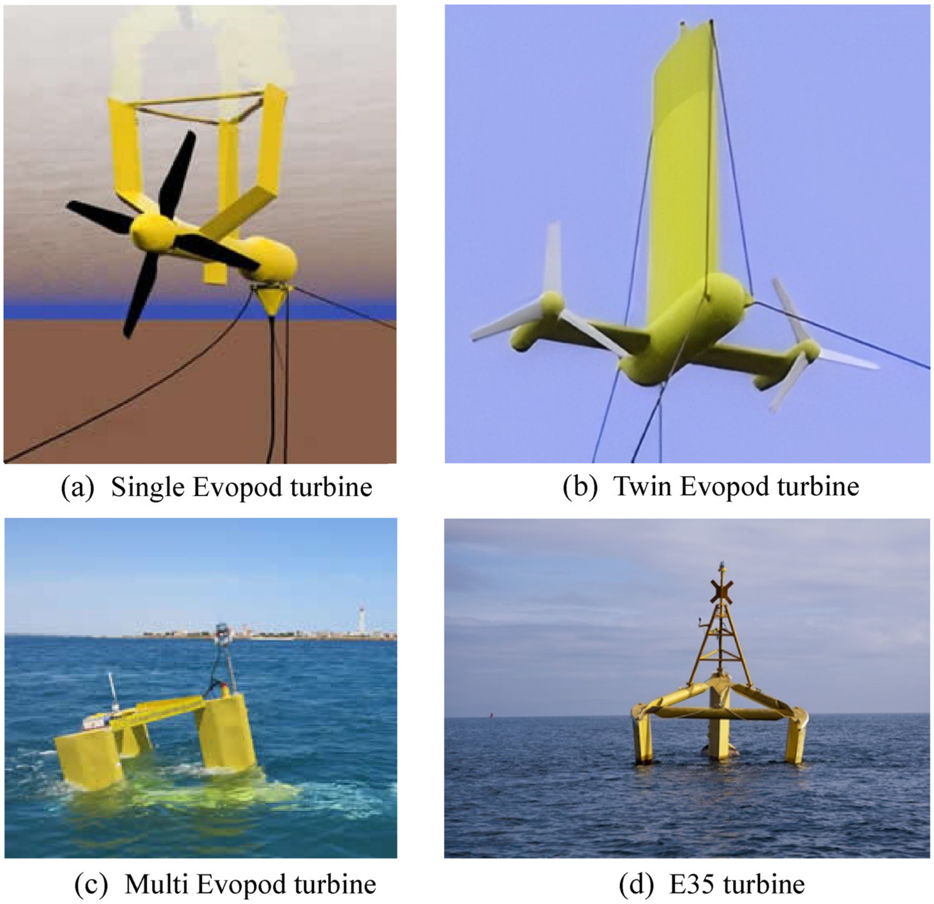


FIG. 22. Oceanflow tidal device.[59] (From Ocean Flow Energy, 2025, https://www.oceanflowenergy.com/.)

FIG. 23. Tocardo tidal device.[61–63] [(a) From Offshore-Energy, 2025, https://www.offshore-energy.biz/; (b) Maritime Cyprus, 2025, https://maritimecyprus.com/; and (c) Tocardo, 2025, https://tocardo.com/.]

To further improve techno-economic efficiency, Tocardo launched the T-2si series in 2020. This generation achieved 95% component commonality through a standardized design approach [Fig. 23(c)]. Featuring a modular design, a rapid innovation curve, and scalable technology, the T-2si series resulted in the development of the most competitive 600 kW medium-scale tidal turbine available at the time.

h. ZJU tidal current energy device (China). In 2006, Zhejiang University (ZJU) developed China's first horizontal-axis tidal current energy device, known as the "Underwater Windmill." Operating on a mechanical transmission principle, the device successfully completed its initial sea trial power generation test in Zhoushan, Zhejiang Province. In 2008, ZJU further advanced its research with the development and successful sea trial of a semi-direct-drive tidal current energy device prototype. By 2010, the university had introduced the country's first hydraulically driven, pitch-regulated tidal current energy generator set.

Building on years of sustained research and development, ZJU developed a 60-kW horizontal-axis semi-direct-drive tidal current energy generator set, along with a dedicated marine test platform.[64] This system has undergone multiple multi-year testing cycles in the Zhoushan Sea [Fig. 24(a)]. Since its commissioning in May 2014, the system has achieved several national records, including the highest

(a) 60 kW Floating tidal device (b) 650kW Floating tidal device

FIG. 24. ZJU tidal device.[65] (From Zhejiang University, 2025, https://www.zju.edu.cn/.).

single-unit daily average power output, the longest continuous operational duration, and the largest cumulative power generation capacity among all tidal current energy devices in China. Additionally, it is also the first tidal current energy system in China to undergo independent third-party testing and validation.

Recently, a 650-kW high-aspect-ratio, high-efficiency horizontal-axis tidal current energy generator set, developed by an interdisciplinary team from ZJU's Ocean College and the School of Mechanical Engineering, was successfully connected to the grid at the Zhoushan Ocean Energy Experimental Power Station [Fig. 24(b)]. This system currently represents the largest single-unit tidal current energy device in China, and its successful grid connection marks a major breakthrough in the country's ocean energy technology.

i. HEU tidal current energy device (China). Harbin Engineering University (HEU) in China successfully developed the "Haineng II" tidal current energy device, a floating system designed for efficient marine energy conversion (Fig. 25). The device features a two-bladed horizontal-axis, pitch-regulated, direct-drive power generation system with a total installed capacity of 200 kW. It employs a four mooring system arranged in a "center" configuration, supporting a floating platform that houses two 100 kW generators and includes a liftable maintenance platform for ease of access and servicing.[66] In June 2013, the Haineng II device was successfully deployed in the waters near Zhaitang Island, Qingdao, transmitting electricity via a 1-km submarine cable to the central control room of the 500 kW island multi-energy complementary independent power system. The successful implementation of this demonstration project validates the technical feasibility of floating tidal current energy systems and provides an innovative solution for renewable energy generation on remote islands.

j. OUC tidal current energy device (China). In June 2015, OUC developed the axial tidal current energy device "Haichuan," which was successfully installed and operated in the waterway near Zhaitang Island, Qingdao.[68] The device features a floating carrier platform integrated with a gantry-lift support mechanism, providing reliable positioning and operational stability while enhancing mobility. This design facilitates towing, relocation, recovery, and routine maintenance. Haichuan incorporates variable-pitch blade technology and a maximum power point tracking control system, allowing for low-flow startup and high-efficiency energy conversion. An underwater real-time monitoring system continuously tracks the device's operational status and enables precise variable-pitch control, ensuring safe and stable performances. Additionally, a remote data acquisition and control system, based on General Packet Radio Service technology, enhances system automation and intelligence. The device has an installed capacity of 20 kW and has demonstrated year-round continuous operation. Field test data indicate an impressive turbine energy capture coefficient of 43.3% and an overall system efficiency of 35.2%, highlighting its strong technical performance and reliability.

FIG. 25. HEU tidal device (Haineng-II).[67] (From College of Shipbuilding Engineering, Harbin Engineering University, 2025, https://sec.hrbeu.edu.cn/.)

B. Vertical axis tidal turbines

1. Pile structure

a. Atlantis tidal current energy device (Singapore). In 2008, Atlantis, the Singapore-based company Atlantis developed the "Nereus" series of vertical-axis tidal current turbines, which includes the 150 kW AN-150 and 400 kW AN-400 models[26] (Fig. 26). This series features a bottom-mounted vertical-axis sail-blade design. In May 2008, Atlantis replaced its initial demonstration device, the Aquanator, with the AN-150 turbine, marking a significant upgrade in design and performance. Subsequently, in July 2008, the company conducted high-seas towing tests of the AN-400 turbine. Test results indicated a record-breaking power output from the AN-400, along with notable improvements in waterline efficiency.

b. EC-OG tidal current energy device (the United Kingdom). EC-OG, a UK-based company with a multidisciplinary technical team, entered into a cooperative agreement with EMEC to conduct testing of its Subsea Power Hub (SPH) system[70] (Fig. 27). In April 2017, the company conducted the first offshore test of a full-scale SPH system at EMEC's Shapinsay Sound Proving Ground, marking a significant technological advancement in subsea energy systems.

The vertical-axis underwater power system developed by EC-OG is designed to extend the operational range of subsea control modules, offering a novel solution for autonomous underwater energy supply. The SPH system represents an innovative hybrid energy conversion and storage platform, integrating real-time charging capabilities with onboard energy storage. In November 2017, the system successfully completed offshore validation trials. The SPH configuration consists of three VATTs, each delivering approximately 500 W of power. The system utilizes a direct-drive generator integrated with an energy storage battery, enabling reliable energy delivery in variable marine conditions.[71] The entire system is housed within a compact trawl frame, occupying a footprint of less than 7 m^2 and standing at a height of just 2 m.

c. LHD tidal current energy device (China). In July 2016, China successfully deployed the first 1 MW unit of the LHD marine tidal current energy project in the waters off Zhoushan, which subsequently began power generation.[72] The system employs a resistance-type VATT mounted on a platform secured to the seabed with embedded rock piles, enabling it to withstand extreme marine meteorological and hydrological conditions. The unit was successfully connected to the grid in August 2016 and has been in continuous grid-connected operation since May 2017. By December 2020, the turbine had achieved over 43 months of continuous operation, with a cumulative power generation exceeding 2×10^6 kWh.

FIG. 26. Atlantis tidal device.[69] (From Das Buch der Synergie, 2025, https://www.buch-der-synergie.de/.)

FIG. 27. EC-OG SPH tidal device.[48] (From European Marine Energy Center, 2025, https://www.emec.org.uk/.)

Building on the success of the project's first phase, the LHD initiative deployed its second-generation 400 kW and third-generation 300 kW units in November and December 2018, respectively. The project currently operates four turbine modules, integrating internationally recognized tidal current energy power generation technologies. With a total installed capacity of 1.7 MW, the LHD project ranks among the global leaders in both installed capacity and technological innovation.

The outstanding performance of the LHD project during its extended operation period highlights China's progress in marine renewable energy development. In 2021, the project reached a new milestone with the successful deployment and grid connection of the fourth-generation 1.6 MW unit, Endeavor (Fig. 28). This marked a significant breakthrough in megawatt-scale tidal current energy technology and established a solid foundation for the future large-scale commercial deployment of tidal current energy systems in China.

2. Gravity structure

a. Blue Energy tidal current energy device (Canada) Blue Energy, a Canadian company, has successfully developed a low-head turbine

FIG. 28. LHD tidal device.[73] (From Daishan, 2025, https://dsnews.zjol.com.cn/.)

known as the Davis Hydro turbine[74] (Fig. 29). The technical foundation of this device is based on the vertical-axis suspended chain wind turbine design originally proposed by Darrieus in 1931. As a vertical-axis, gravity-based tidal current energy device, the Davis turbine captures hydrokinetic energy from riverine and tidal flows to generate clean, renewable electricity. The turbine's core structure consists of four fixed hydrofoil blades, arranged symmetrically around a vertical axis. These blades transfer mechanical energy to a generator through a rigid drive shaft.

3. Floating structure

a. HydroQuest tidal current energy device (the United Kingdom) In 2019, the 1-MW OceanQuest tidal current energy device, developed by the UK-based company HydroQuest, successfully completed technical validation and was deployed at the Paimpol-Bréha site operated by Electricité de France[76] (Fig. 30). This demonstration project forms part of the Renewable Energy Technology Innovation Funding Program, led by the French Environment and Energy Management Agency. HydroQuest has since developed a second-generation 2.5 MW VATT that demonstrates a 23% improvement in energy conversion efficiency and a 40% increase in mean time between failures, achieved through lightweight structural design and hydrodynamic optimization. With its enhanced reliability and performance, the

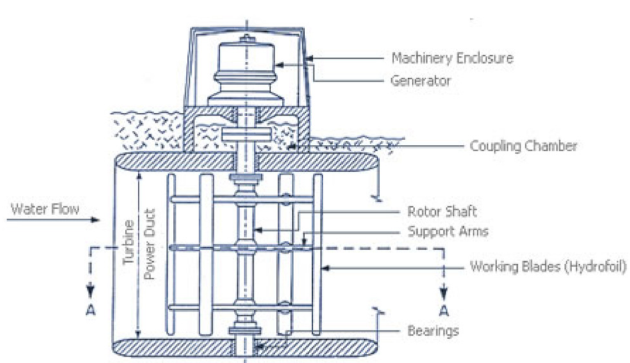


FIG. 29. Blue Energy Davis Hydro tidal device.[75] (From Blue Energy Canada, 2025, https://www.bluenergy.com/.)

FIG. 30. HydroQuest OceanQuest Hydro tidal device.[77] (From Tiger, 2025, https://interregtiger.com/.)

turbine's stand-alone capacity ranks among the highest in the global tidal current energy sector.

As part of the upcoming Flowatt project, scheduled for deployment in 2027, six OceanQuest turbines will be installed to form a 17 MW power plant at Le Raz Blanchard, with an expected operational lifespan of 20 years. Through this project, HydroQuest aims to further advance and validate subsea tidal current energy technology, reinforcing the OceanQuest platform's competitive position in the rapidly evolving global tidal current energy market.

b. Design Pro tidal current energy device (Ireland) The HydroKinetic tidal current energy device, developed by Design Pro Renewables in Ireland,[78] features two VATTs mounted on either side of a flow-accelerating structure known as a "blunt body" (Fig. 31). The aerodynamic design of the blunt body, combined with an integrated blade pitch control system, accelerates the water flow through the device, thereby enhancing power output and energy conversion efficiency. The device harnesses the kinetic energy of flowing water in rivers and estuaries by exploiting the natural interaction between water

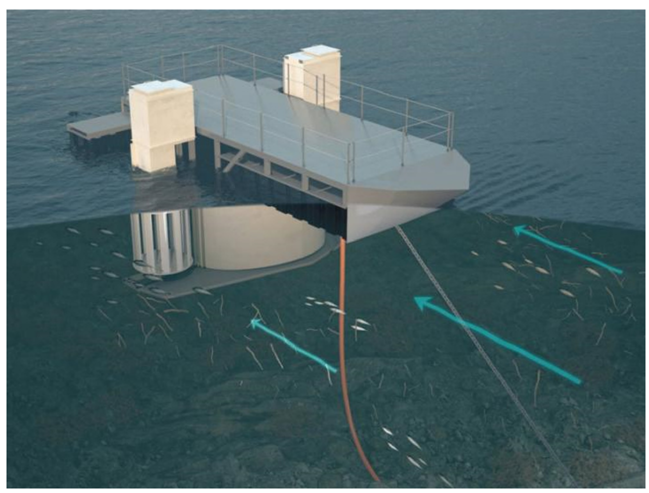


FIG. 31. Design Pro HydroKinetic Hydro tidal device.[79] (From DesignPro Renewables, 2025, https://designprorenewables.com/.)

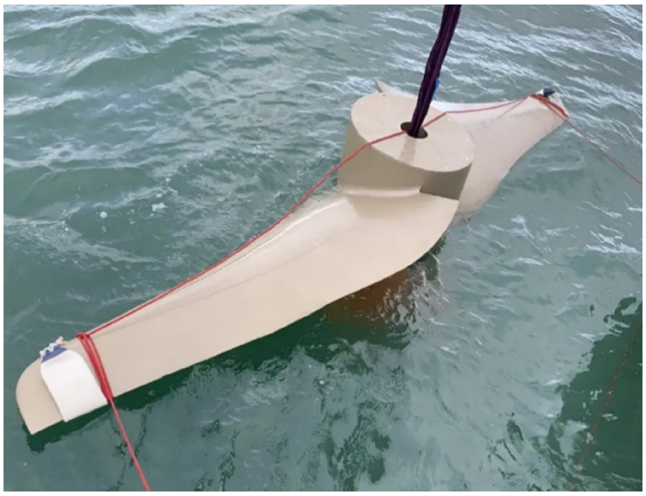


FIG. 32. Flex Marine Power SW tidal device.[81] (From Crowdcube, 2025, https://www.crowdcube.com/.)

currents and physical obstructions. The carefully engineered blunt body compresses and redirects the flow, increasing its velocity as it passes through the turbine zone, which significantly improves performance under variable flow conditions. The compact and modular design of the device allows for rapid and cost-effective deployment. The 25 kW system can be installed in as little as three hours without the need for extensive civil works or specialized infrastructure, making it particularly suitable for remote or off-grid locations with limited access to construction resources.

c. Flex tidal current energy device (the United Kingdom) The floating platform design for the Flex tidal current energy device enhances its operational flexibility, allowing for repositioning as needed and enabling direct access to the turbine for maintenance without requiring full system disassembly[80] (Fig. 32). The core structure comprises double hollow glass fiber blades, connected to a sealed dry cabin via a high-strength steel hub. The cabin houses the integrated generator set and an intelligent monitoring system, facilitating real-time performance tracking and diagnostics. The device is connected to the subsea mooring system through a tubular mooring frame, forming a stable dynamic response structure capable of withstanding variable marine conditions. The mooring system comprises three key components: a subsea mooring assembly, a steel buoyant module, and a buoyant rotor drive assembly. The turbine is installed in a cantilevered configuration at the end of the tubular support, featuring a 5-m rotor diameter and a swept area of 20 m^2.

In 2023, the SW2 tidal current energy converter, based on this platform design, successfully passed the technical certification process in accordance with IEC TS 62600-4. It became the first tidal current energy device globally to receive an IECRE feasibility statement from issue by Lloyd's Register.

d. HEU tidal current energy device (China) In 2002, HEU serving as the technical lead, developed China's first 70 kW floating tidal current energy experimental power plant, named "Wanxiang I," which was built in Zhejiang Province[82] [Fig. 33(a)].

In August 2012, HEU collaborated with four industrial partners to develop the "Haineng I" vertical-axis tidal current energy device [Fig. 33(b)].[83] The system features a floating design and an installed capacity of 300 kW, with a VATT suspended beneath a catamaran-style platform and directly connected to a direct-drive generator. The device was installed and commenced test operations in Zhejiang Province.

Building on this foundation, the "Haineng III" device incorporates a floating double-layer, configuration with two-bladed vertical-axis turbines, each supporting independent 300 kW generating units[84] [Fig. 33(c)]. The turbine transmits mechanical power to the generators via gearboxes, enabling independent operation. The Haineng III device was successfully installed at sea in August 2013.

C. Oscillating hydrofoil

1. EB tidal current energy device (the United Kingdom)

In 2002, the UK-based company Engineering Business (EB),[87] completed the development of the first oscillating hydrofoil tidal current energy device, known as Stingray, with a rated capacity of 150 kW (Fig. 34). The Stingray device utilizes a gravity-based foundation and was installed in the Shetland Islands, Scotland, in 2003. During operational testing, the device demonstrated a stable average power output of 117.5 kW at under rated flow conditions.

2. Pulse tidal current energy device (the United Kingdom)

The Pulse100 device, developed by Pulse Tidal in the UK, employs a dual oscillating hydrofoil system, incorporating upper and

FIG. 33. HEU tidal device.[67,85,86] [(a) From College of Shipbuilding Engineering, Harbin Engineering University, 2025, https://sec.hrbeu.edu.cn/; (b) Harbin Engineering University Engineering Gong Xue News, 2025, https://news.hrbeu.edu.cn/; and (c) Membrane Industry Association of China, 2025, http://www.membranes.com.cn/.]

FIG. 34. Engineering Business Stingray tidal device.[88] (From Tethys, 2025, https://tethys.pnnl.gov/.)

lower hydrofoils for tidal current energy conversion[89] (Fig. 35). The device primarily consists of a structural frame, rocker arms, and airfoil blades. The base of each rocker arm is hinged to the vertical support column, while its distal end is connected to a horizontal wing plate. As the tidal current flows around the wing plates, hydrodynamic lift is generated, inducing an oscillatory up-and-down motion of the wing plate and rocker arm around the pivot point. This oscillation drives a hydraulic cylinder located between the rocker arm and the support column, converting mechanical motion into high-pressure hydraulic energy. The hydraulic energy is then used to power a hydraulic motor, which drives the generator to produce electricity. The Pulse100 device is capable of operating under bidirectional tidal flow conditions, offering versatility in varying marine environments. Its hydraulic system provides benefits in terms of energy stabilization and short-term storage, while its simple blade design facilitates manufacturability. However, the system's monitoring and control components are relatively complex. In September 2009, a 100-kW prototype of the Pulse100 device was deployed in the River Humber for operational testing, achieving over two years of continuous operation. Despite the success in deployment, the overall energy conversion efficiency was limited to approximately 20%, primarily due to energy losses in hydraulic transmission and mechanical drive system. In response, Pulse Tidal is conducting further optimization studies with the goal of increasing the system's efficiency to 30%.

FIG. 35. Pulse Tidal Pulse100 tidal device.[90] (From Rotherham Business News, 2025, https://www.rothbiz.co.uk/.)

3. EEL tidal current energy device (France)

The French company EEL Energy has been actively testing its EEL tidal current energy device since mid-2023[91] (Fig. 36). The design is inspired by the wave-film principle and mimics the undulatory motion of fish to generate energy. It is available in various power capacities, including 5, 30–50, 100, and 1000 kW, making it adaptable to a wide range of deployment scales and marine environments. The EEL device utilizes a flexible, oscillating membrane, with its entire surface acting as an energy-harvesting sensor to maximize energy capture efficiency. Theoretically, the design allows for the capture of up to 100% of the available kinetic energy in the flow, according to fluid dynamic principles.

EEL Energy is currently developing a 1-MW power generation set, composed of two 500 kW units, with full-scale field testing planned for 2026.

4. ResHydro tidal current energy device (the United States)

In 2013, the US-based company ResHydro established a research facility in Glasgow, Scotland, with the support of a $153 000 grant from Scottish Enterprise[92] (Fig. 37). The company is currently collaborating

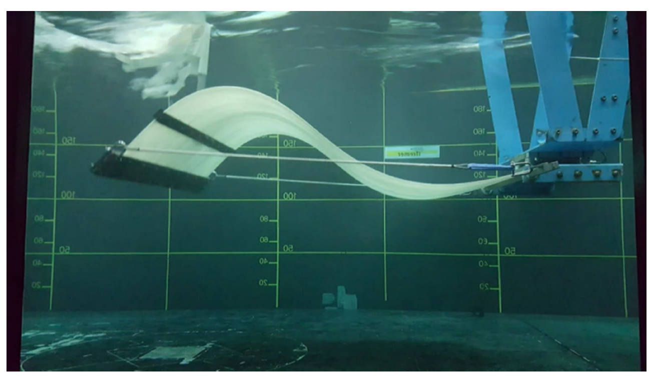


FIG. 36. EEL Energy EEL tidal device.[61] (From Offshore-Energy, 2025, https://www.offshore-energy.biz/.)

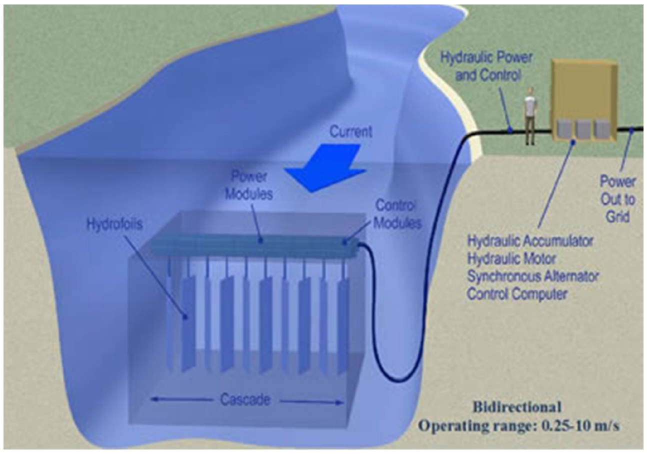


FIG. 37. ResHydro HCR tidal device.[93] (From Renewable Energy World, 2025, https://www.renewableenergyworld.com/.)

with the University of Strathclyde on the development of an innovative tidal current energy system known as the Hydrofoil Cascade Resonator. Distinct from conventional turbine-based power generation systems, the ResHydro device does not rely on rotational motion to drive electricity generation. Instead, it utilizes water currents to induce oscillatory vibrations—a phenomenon the company terms "flutter."

D. Ducted tidal turbine

1. SAE tidal current energy device (Singapore)

Since its inception, the Singaporean-based company SAE Renewables has been actively engaged in the development and testing of advanced tidal current turbines. In 2008, the company successfully introduced the "Solon" AS series of HATTs.[26] Designed for optimal performance at an operational flow velocity of 2.6 m/s, the AS series includes turbine models with rated capacities of 160 kW, 500 kW, and 1 MW. The AS series have undergone extensive testing under various environmental conditions, demonstrating strong adaptability and performance reliability. The results confirm the AS series' technical feasibility and commercial potential in tidal current energy applications.

2. OpenHydro tidal current energy device (Canada)

The Open-Center turbine, developed by OpenHydro, an Ireland-based company, was deployed as part of CSTV project for tidal current energy development, implemented by the Canadian Tidal Energy Development Center[94] (Fig. 38). The turbine features a closed-tip (Venturi-style) structure and employs a horizontal-axis design equipped with ten airfoils. Each airfoil measures 4.8 m in length and 2.4 m in width and is constructed from glass-reinforced plastic. The blade thickness tapers from 21 cm at the root to 1.5 cm at the tip, optimizing hydrodynamic performance. The turbine is supported by a submerged triangular gravity structure and is deployed directly on the seabed without requiring drilling or substrate preparation. OpenHydro's Generation 7 turbine, with a 6-m rotor diameter, was installed in April 2014 and has since accumulated over 10 000 h of operational runtime.

FIG. 38. OpenHydro Open-Center tidal device.[48] (From European Marine Energy Center, 2025, https://www.emec.org.uk/.)

3. DHV tidal current energy device (Australia)

The Davidson Hill Venturi (DHV) turbine is a VATT incorporating a distinctive Venturi design, developed by DHV Turbines in Australia[95] (Fig. 39). This innovative configuration enables superior performance in low-flow environments, with power output enhancement factors of up to threefold, corresponding to a 300% increase in potential power output. This remarkable performance of the DHV turbine is primarily attributed to the unique effect of the slat diffuser, which effectively concentrates water flow onto the turbine blades. During high-speed flow testing at velocities of 6 m/s or higher, the device achieved an enhancement coefficient of 2.4, further validating its hydrodynamic efficiency and design advantages.

4. HEU tidal current energy device (China)

In December 2005, HEU constructed the 40 kW "Wanxiang II" tidal current power plant in Zhejiang Province[97] [Fig. 40(a)]. The plant

FIG. 39. DHV tidal device.[96] (From Sail-World, 2025, https://www.sail-world.com/.)

(a) Wanxiang II (b) Haiming I

FIG. 40. HEU tidal device (Wanxiang II and Haiming I).[67] (From College of Shipbuilding Engineering, Harbin Engineering University, 2025, https://sec.hrbeu.edu.cn/.)

utilizes a gravity-based support structure, and its power generation system is described as a gravity friction-fixed dual-rotor current-guided enhanced tidal current stand-alone energy system. The primary components include a ducted box-type structural carrier for flow enhancement, a twin-rotor turbine, a mechanical speed-increasing mechanism, an electrical control system, and auxiliary subsystems. The main shaft of the twin-rotor hub-and-spoke turbine is supported at its upper end by a float box located above the deflector, and at its lower end by a caisson, further stabilized by eight supporting legs, ensuring structural integrity and operational stability in dynamic marine conditions.

The subsequent development, the "Haiming I" tidal current energy device, also designed by HEU, represents the first long-term demonstration of a gravity-based, horizontal-axis ducted tidal current stand-alone power generation system in China[98] [Fig. 40(b)]. The device adopts a ducted structural design and has a rated power output of 10 kW. In September 2011, "Haiming I" underwent successful sea trials in Zhejiang Province, supplying electricity via a submarine cable to the Xianzhou Bridge lighthouse, where it was used for lighting and heating.

E. Other forms

1. SRI tidal current energy device (the United States)

The SRI Manta, a US-based research organization, has developed a tidal kite energy system designed to harness the energy of ocean currents, based on principles similar to those of an airborne kite[99] (Fig. 41). The device, named "Manta," draws inspiration from the elegant propulsion of manta rays. The power conversion system is relatively simple and operates through a pumping action—alternating line retraction and release—to convert the kite's motion into electrical energy. Unlike conventional systems that depend on rigid, fixed-axis rotating turbines, the Manta provides a safer, more environmentally friendly, and community-compatible method of tidal power generation.

When deployed, the tidal kite pulls against its tether, driving a generator to produce electricity from tidal current energy. The system's trajectory can be dynamically adjusted to accommodate seasonal variations, enabling the kite to avoid storms, vessels, and marine animals. Moreover, due to the absence of exposed rotating components, the system poses minimal risk to marine life, offering a promising alternative to traditional tidal current energy technologies.

2. SeaPower tidal Current Energy Device (Italy)

In 2020, the Italian company SeaPower Scrl developed the GEMSTAR tidal current energy device,[101] a tidal kite system in particular, designed to generate electricity from slow-moving marine currents (Fig. 42). The GEMSTAR configuration features two counter-rotating turbines mounted on the either side of a submerged hydrodynamic structure, which also houses the system's electronic and control components. The design incorporates tail planes that provide both longitudinal and transverse stability, allowing the system to maintain optimal flow alignment even under bidirectional tidal conditions. During operation, the GEMSTAR device behaves similarly to a submarine kite, remaining in a relatively stable position due to the equilibrium between the hydrodynamic thrust acting on the turbines and structures and the buoyancy force generated by the system's hollow section.

3. SeaQurent tidal current energy device (Netherlands)

In 2023, the Dutch company SeaQurrent conducted initial tests of its TidalKite tidal current energy device,[103] laying the groundwork

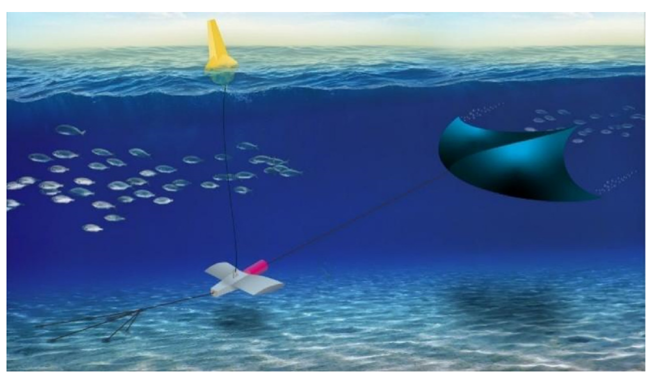


FIG. 41. SRI Manta tidal device.[100] (From SRI International, 2025, https://www.sri.com/.)

FIG. 42. SeaPower Scrl GEMSTAR tidal device.[102] (From GEMSTAR, 2025, https://www.gemstar.it/.)

FIG. 43. SeaQurent TidalKite tidal device.[61] (From Offshore-Energy, 2025, https://www.offshore-energy.biz/.)

for a future offshore deployment in the Netherlands (Fig. 43). The TidalKite system employs tidal kite technology to harness kinetic energy from marine currents, operating along a predefined, stable figure-eight trajectory to maximize energy capture. The entire system is autonomously controlled by a state-of-the-art remote monitoring and control platform, which continuously optimizes performance and ensures operational safety. During operation, the tensile force generated by the moving kite drives the flow of a biodegradable working fluid, which is then converted into rotational mechanical energy to power a generator and produce electricity. The generated power is transmitted to shore via submarine cables and integrated into the existing electrical grid. The TidalKite system is designed for broad adaptability across various tidal environments and can be towed, deployed, and retrieved using standardized, commercially available vessels and equipment. This modular and accessible deployment strategy contributes to a reduction in the LCOE, enhancing the system's commercial viability.

4. Minesto tidal current energy device (Sweden)

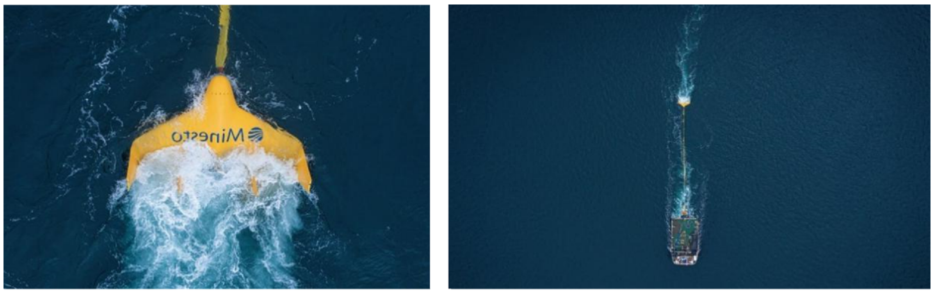
The Dragon 4 system, developed by the Swedish company Minesto, utilizes a tidal kite design and became operational in 2022[104] [Fig. 44(a)]. With a rated capacity of 100 kW, the system is in particular, designed for microgrid applications, particularly in remote or island communities. Its streamlined design and lightweight structure facilitate easy deployment and maintenance. Installation is achieved by gently lifting and maneuvering the system via a tether, allowing for a rapid and efficient setup process. The Dragon 4 can be quickly deployed with minimal labor, enabling fast commencement of power generation.

Minesto's most recent advancement, the Dragon 12, represents a significant scale-up, with a rated capacity of 1.2 MW, tailored for utility-scale energy generation [Fig. 44(b)]. The first Dragon 12 unit was commissioned in February 2024 and has successfully delivered its initial power output to the Faroese electrical grid.

5. Flumill tidal current energy device (Norway)

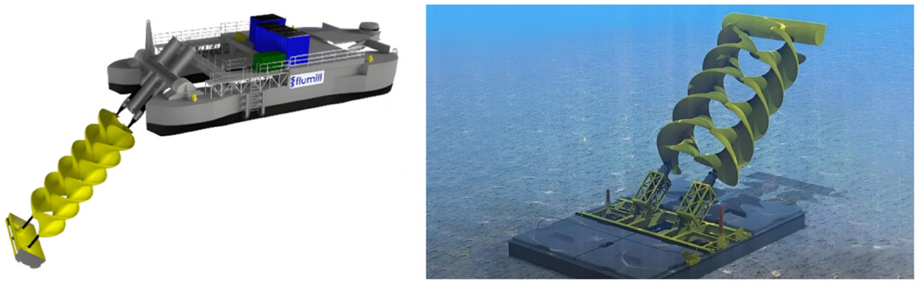
The Flumill Floating System, developed by the Norwegian company Flumill, consists of a twin-turbine assembly mounted on a catamaran floating platform[106] [Fig. 45(a)]. The system's power output is primarily determined by the turbine diameter and is based on the Archimedes screw principle, utilizing a double-screw configuration.

Flumill's second-generation design, the Flumill Seabed System, features a similar twin-turbine assembly mounted on a bottom-fixed support frame [Fig. 45(b)]. The output of this system is a function of both turbine diameter and local flow velocity. The structural support is constructed from steel frameworks, which are anchored to the seabed using either pre-bored piles or gravity-based foundations. Each unit



(a) Dragon 4 tidal kite (b) Dragon 12 tidal kite

FIG. 44. Minesto tidal device.[105] (From Minesto, 2025, https://minesto.com/.)



(a) Flumill Floating System (b) Flumill Seabed System

FIG. 45. Flumill tidal device.[107] (From Flumill, 2025, https://flumill.com/.)

includes two counter-rotating turbines with diameters ranging from 30 to 40 m, mounted directly on the seabed. The opposite rotational directions contribute to improved system stability during operation. As tidal water flows through the system, it rotates the screw shape turbines, which in turn directly drive a gearless permanent magnet generator.

6. Jupiter tidal current energy device (Canada)

The Jupiter Hydro technology, developed by Jupiter Hydro Inc. in Canada, has undergone successful validation and has been confirmed to pose no harm to fish during operation[108] (Fig. 46). Building on these findings, the company adopted the 2500-year-old Archimedes screw principle as the foundation of its energy conversion design, integrated into a modern floating platform system. The construction and deployment of the full-scale device are scheduled for completion by fall 2025.

7. Kepler tidal current energy device (the United Kingdom)

In 2018, the UK-based company Kepler Energy developed an innovative tidal turbine system[110] (Fig. 47). Test results have confirmed that its potential energy conversion efficiency significantly exceeds that of conventional axial-flow turbine designs. Kepler's technology is based on transverse HATTs configuration, which can be deployed either independently or in array formations, functioning similarly to a tidal fence. Experimental validation has demonstrated that the power output of the system is multiple times greater than that of a traditional turbine when deployed in the same location. The core generator assembly consists of two turbines linked to a central direct-drive generator, supported by just four bearings and three foundation structures offering both mechanical simplicity and reduced structural weight compared to conventional propeller-based systems. The transverse horizontal-axis configuration also provides bidirectional operational efficiency, eliminating the need for yaw mechanisms or active alignment when the tidal flow direction reverses.

F. Technical comparison

A comparative analysis of the performance characteristics of various tidal current energy conversion technologies is presented in Table I.

Among the various tidal current energy technologies, axial turbines are currently the most widely deployed and internationally favored, largely due to their hydrodynamic stability, high energy conversion efficiency, and adaptability across a range of flow conditions. Conversely, emerging technologies—such as oscillating hydrofoils, ducted turbines, tidal kites, and Archimedes screws—are progressing from theoretical exploration to practical implementation. Despite their unique design features and potential benefits (e.g., reduced environmental impact, adaptability to diverse flow regimes), these systems still face notable technical limitations and require further research and optimization before achieving widespread commercial adoption.[112–120]

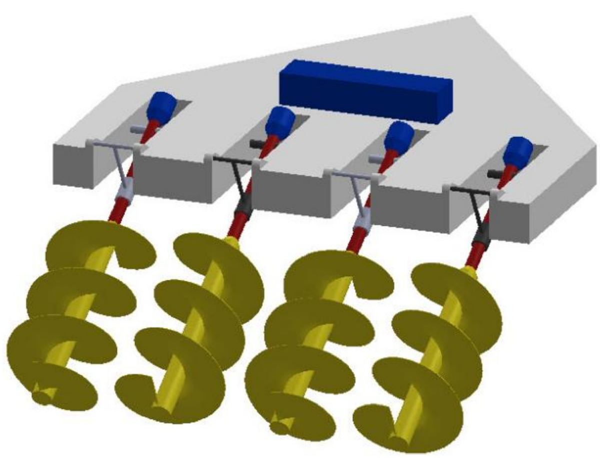


FIG. 46. Jupiter tidal device.[109] (From Jupiter Hydro, 2025, https://jupiterhydro.com/.)

FIG. 47. Kepler Energy THAWT tidal device.[111] (From New Atlas, 2025, https://new-atlas.com/.)

TABLE I. Comparative performance characteristics of seven tidal turbine types.

Device type	Operating principle	Advantages	Disadvantages
Horizontal axis turbine (HATT)	The rotor shaft aligns parallel to the water flow direction	- High stability - High energy conversion efficiency- Broad flow rate adaptability	- Complex structure design - Yaw control - Significant tip losses - High leaf tip loss - High noise emission
Vertical axis turbine (VATT)	Rotor shaft perpendicular to the direction of water flow	- Simple design - Easy installation - Low sealing requirements - Suitable for multidirectional flows - Low noise operation	- Lower energy conversion efficiency - Requires external starting torque - Limited high-flow performance - Low output torque
Oscillating hydrofoil	Reciprocating motion of a lifting body	- Slow motion - Greater hydrodynamic freedom - Low noise level	- Requires precise angle adjustment - Susceptible to collisions
Ducted turbine	Flow acceleration achieved through a duct structure	- High energy conversion efficiency - Economical construction	- Performance characteristics similar to conventional turbines
Tidal kite	Moves in a figure-8 trajectory with mooring	- Effective in low-flow velocity environments - Adaptable to changing flow conditions	- High collision risk - Potential high-frequency noise impact
Archimedes screw	Utilizing spiral blades to convert kinetic energy into rotational energy	- Low risk of collision - Low noise emission - Environmentally friendly	- Limited to specific application scenarios - Low technological maturity
Tidal fence	Harnesses energy from tidal head differences through flow-constrained pathways	- Stable and consistent power output - Proven, mature technology	- High capital and construction costs - Potential disruption to marine life migration and ecosystems

IV. ADVANCES IN HYDRODYNAMIC ANALYSIS METHODS FOR OCEAN TIDAL CURRENT ENERGY DEVICES

Hydrodynamic analysis plays a critical role in evaluating the fluid-structure interaction characteristics of tidal current energy devices, specifically within the dynamic conditions of a marine environment. This analysis investigates how fluid forces influence structural performance and, consequently, how these interactions affect the overall energy conversion efficiency and operational stability of the device. In this study, a tidal current energy device is conceptually divided into two primary subsystems: the turbine and the support structure. Each subsystem exhibits distinct hydrodynamic characteristics that are essential to the energy capture and conversion process. The turbine serves as the core energy-harvesting component, converting the kinetic energy of flowing water into mechanical energy via rotational motion. In contrast, the support structure maintains the positional stability integrity of the turbine system, ensuring efficient operation in varying tidal conditions. To conduct a comprehensive hydrodynamic assessment, it is essential to investigate not only the individual fluid dynamic behavior of the turbine and support structure but also their mutual coupling effects.

A. Turbine hydrodynamic analysis

1. BEM method

The BEM method is widely used as an analytical approach for evaluating the hydrodynamic performance of tidal current energy turbines. This approach method combines momentum theory with blade element theory to characterize the interaction between turbine blades and the incoming flow by incorporating induced velocity factors. The core principle of BEM involves discretizing the rotor into multiple radial blade elements, independently calculating the lift and drag forces on each element, and subsequently determining the overall thrust and power output, accounting for rotationally induced velocity (Fig. 48).

BEM is known for its computational efficiency and is particularly well-suited for performance prediction and parametric optimization in the preliminary design phase. For example, Perez *et al.*[121] applied BEM in conjunction with high-frequency Acoustic Doppler Current Profiler data to investigate the effects of turbulence intensity on turbine thrust and power. Their findings showed that while turbulence led to increased load fluctuations, BEM remains effective in predicting the impact on power and thrust coefficients. Similarly, Sadeqi *et al.*[122] utilized BEM, incorporating JavaFoil to calculate lift and drag coefficients and applying Prandtl tip loss correction to improve computational accuracy. Their analysis under various flow conditions confirmed

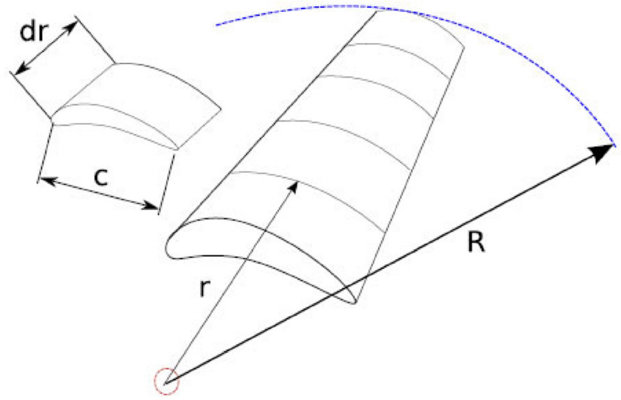


FIG. 48. Hydrodynamic analysis of turbines with BEM method.[124] (From University of Strathclyde, 2025, https://www.strath.ac.uk/.)

BEM's applicability, demonstrating that optimizing rotational speed significantly enhances power output. Fu *et al.*[123] expanded the BEM framework to account for wave-current interactions, introducing blockage corrections to improve prediction accuracy under non-uniform inflow conditions. The high accuracy of BEM in uniform flow fields has been experimentally validated.

However, the BEM method is subjected to several inherent limitations. It is based on the assumption of a homogeneous and steady flow field and does not capture complex transient dynamics or large-scale turbulent structures. Furthermore, turbine-flow interactions are represented only indirectly through induced velocity coefficients, making the method unsuitable for detailed flow field analysis.

2. Vortex method

The BEM method, while efficient, has inherent limitations in capturing the unsteady evolution of wake structures and the detailed flow behavior near blade surfaces. These limitations hinder its ability to accurately predict vortex shedding and its impact on turbine hydrodynamic performance. To address these challenges, the vortex method, based on potential flow theory, has been developed as a more advanced alternative. The vortex method determines hydrodynamic characteristics by tracking the motion of shedding vortices to calculate the pressure distribution and induced velocities on the blade surface. This technique enables high-fidelity prediction of wake dynamics, unsteady hydrodynamic loading, and the influence of blade motion on the surrounding flow field. Moreover, its relatively low computational cost—especially in low Reynolds number conditions—makes it particularly attractive for turbine design and analysis. Based on their modeling approaches, vortex methods can be categorized into: discrete vortex methods, vortex panel methods, finite vortex methods, and free vortex methods.

In the context of tidal current turbine hydrodynamics, Ji *et al.*[125] developed a CFD model incorporating coupled longitudinal oscillation based on the improved delayed detached eddy simulation method. This model effectively predicts the evolution of blade wake and the associated hydrodynamic loads. The results demonstrated that both the frequency and amplitude of longitudinal oscillation have a significant impact on energy utilization efficiency and wake oscillation behavior. Building on this, Liu *et al.*[126] applied a similar turbulence modeling approach to investigate the hydrodynamic response of tidal turbine blades under yaw conditions. Their findings indicate that the model successfully captures vortex shedding and wake asymmetry induced by yaw. It also clearly reveals how variations in yaw angle affect blade thrust and power output. Furthermore, Wang *et al.*[127] employed the vortex panel method to study a vertical-axis tidal turbine. They established a two-dimensional potential flow model that maintains high computational efficiency while accurately describing the pressure distribution on the blade surface and the influence of vortex shedding on overall turbine performance.

Despite its strengths, the Vortex Method has limitations under high Reynolds number conditions, as it is inherently based on inviscid flow assumptions. This makes it difficult to account for viscous effects directly. Additionally, its accuracy depends on the distribution of vortex particles and the selection of time step sizes. While the method offers computational efficiency in small-scale or short-duration simulations, it can become computationally intensive for long-term simulations or large-array configurations. For complex three-dimensional flows, model enhancements are required to accurately capture turbulent diffusion and viscous dissipation effects.

3. CFD method

While the vortex method improves the accuracy of wake prediction, it remains constrained by its reliance on potential flow theory, limiting its capacity to fully characterize complex flow phenomena. In contrast, CFD enables high-fidelity numerical simulations of fluid behavior by directly solving the Navier–Stokes (N–S) equations across the entire computational domain. CFD approaches can be classified based on their solution strategies into two main categories: direct numerical simulation (DNS) and indirect numerical simulation, which includes large eddy simulation (LES) and Reynolds-averaged Navier–Stokes (RANS) methods. DNS resolves all scales of turbulence without the use of turbulence models, providing highly accurate results. However, due to its extraordinary computational demands, it is impractical for most engineering-scale turbine simulations. In contrast, indirect numerical methods strike a balance between accuracy and efficiency: LES resolves large-scale turbulent eddies while modeling only the smaller scales, making it particularly suitable for analyzing transient flow characteristics and unsteady vortex dynamics. RANS simplifies the N–S equations through time-averaging, significantly reducing computational cost while capturing mean flow behavior, making it ideal for steady-state analyses and parametric studies.

The CFD method plays an important role in the hydrodynamic analysis of tidal current turbines due to its capability to model complex, three-dimensional flow fields. For instance, Suhri *et al.*[128] applied CFD to investigate the wake interactions within a turbine array, effectively resolving wake recovery patterns and informing optimal array spacing. Mullings and Stallard[129] employed CFD to examine blade-scale flow characteristics, assessing the effects of unsteady turbulence on blade lift and drag forces, and subsequently used the results to refine BEM model predictions, improving agreement with experimental data. Zhang *et al.*[130] combined the LES and RANS methods to analyze the unsteady response of tidal turbine blades under various winglet lengths and inclination angles. Their results showed that a 45° angle winglet inclination improves power efficiency by approximately 11% and significantly reduces load fluctuations caused by blade tip vortices. Similarly, Manolesos *et al.*[131] examined the influence of passive vortex generators on flow separation control in tidal turbine blades. They used wind tunnel experiments and RANS simulations. The study confirmed that CFD techniques can accurately capture near-wall turbulent structures induced by vortex generators, highlighting their potential for blade optimization and flow control in tidal current energy systems.

Despite its advantages, CFD simulations—particularly those involving high-resolution turbulence models such as LES—require substantial computational resources, which can constrain their use in large-scale or long-duration simulations. Additionally, CFD is highly sensitive to factors such as turbulence model selection, mesh quality, boundary condition settings, and domain configuration. Variability across turbulence models may lead to discrepancies in predicted results, highlighting the need for validation against experimental data.

To balance computational accuracy and efficiency, especially for multi-turbine arrays and complex geometries, CFD is often coupled with experimental measurements or integrated with reduced-order models such as BEM. This hybrid approach enhances reliability and supports robust turbine design optimization.

4. ALM

The ALM was developed as a computationally efficient alternative for simulating the hydrodynamic performance of rotating turbines while maintaining a high degree of accuracy. ALM combines the principles of BEM theory with flow field modeling based on the N–S equations, and is commonly used within CFD simulations to approximate the aerodynamic or hydrodynamic effects of a rotor. Rather than resolving the full blade geometry through detailed meshing—which significantly increases computational cost—ALM introduces virtual actuator lines into the computational domain. These lines represent the rotating blades, and the aerodynamic forces generated by the blades are calculated using local flow conditions and blade element theory. The resulting forces are then distributed as body force terms and projected onto the background computational mesh, effectively influencing the flow field without requiring explicit modeling of blade surfaces.[132]

While the ALM offers significant computational advantages, it cannot independently capture the full complexity of turbine hydrodynamics. Therefore, ALM is often integrated with other numerical methods to enhance the accuracy of simulations involving flow structures, wake dynamics, and turbine loading characteristics. Li *et al.*[133] employed ALM in conjunction with the RANS solver to analyze the effects of initial yaw angles, induced frequencies, and amplitudes on the performance of tidal current energy turbines. Their approach successfully captured the high-frequency oscillatory behavior of power and thrust coefficients during yaw-induced motion and clarified the influence of yaw angle on amplitude-averaged hydrodynamic coefficients. In a subsequent study, Li *et al.*[134] further enhanced ALM by integrating it with a sliding mesh technique to simulate forward and transverse rocking motions of a turbine mounted on a floating platform. This method effectively characterized the impact of coupled platform motions on wake vortex structures, revealing how the interplay of forward momentum and transverse rocking motions alter the expansion-contraction oscillation mode of the wake. Deng *et al.*[135] coupled ALM with the immersed boundary method to investigate the influence of turbine-induced rotational hydrodynamics on local seabed scour around the support structure. Their findings demonstrate that turbine rotation intensifies seabed flow velocities, increasing the scour depth near monopile foundations. Notably, the support structure had a greater influence on local scour than the turbine's rotational effects alone. To improve ALM's accuracy under coarse mesh conditions, Stanly *et al.*[136] developed the Filtered Actuator Line Model. This enhanced model addresses the common issue of lift overprediction in standard ALM by optimizing the computation of induced velocities, particularly in the blade tip and root regions. As a result, the Filtered Actuator Line Model provides improved simulation accuracy in under-resolved grid environments, making it suitable for large-scale simulations with limited computational resources.

However, in near-field wake simulations, the ALM may exhibit vortex structure diffusion, limiting its ability to accurately resolve wake dynamics, particularly in regions of intense vortex shedding and interaction. Additionally, standard ALM implementations tend to overpredict lift forces near blade tips when applied on coarse computational meshes. This limitation often necessitates the use of filtering techniques or model corrections to improve computational accuracy.

B. Support structure hydrodynamic analysis

1. Morison formula

The Morison empirical formula, originally proposed by Morison *et al.*, has been widely adopted in hydrodynamic analysis of marine structures. The formulation decomposes the total wave-induced force into two primary components: a drag force associated with fluid viscosity and an inertial force resulting from the acceleration of the surrounding fluid mass.

However, when the support structure undergoes elastic deformation, the traditional Morison formula becomes inadequate for accurately capturing the dynamic characteristics of wave-induced loading. To improve its computational accuracy and predictive capability, researchers have attempted to integrate the Morison formula with complementary modeling approaches. For instance, Jin[137] combined the Morison formula with potential flow theory to analyze the hydrodynamic behavior of horizontal cylinders subjected to regular wave conditions. The study demonstrated that, while the Morison formula provides reasonable estimates of inertial and drag forces, it is limited in accounting for structural elasticity and the resulting dynamic response of deformable structures. Similarly, Zan[138] investigated the applicability of the Morison formula in the context of internal waves and identified notable biases in predicting the temporal evolution of internal wave-induced loads. To enhance the applicability of the Morison formula in performance evaluation and complex flow environments, researchers have recently introduced empirical correction factors, integrated it with potential flow theory, and applied numerical optimization techniques via CFD. These advancements aim to improve its modeling fidelity, enabling more accurate assessments of hydrodynamic loads in nonlinear, unsteady, and deformable system contexts.

2. Analytic method

In contrast to the empirical Morison formula, the analytical method involves the development of physical models to derive explicit expressions for hydrodynamic forces acting on structures subjected to wave loading. By integrating wave theory with fundamental hydrodynamic principles, this approach offers improved theoretical rigor and greater predictive accuracy. The analytical method enables the formulation of well-defined mathematical model, allowing for quantitative analysis, parameter sensitivity studies, and optimization of structural design. Common analytical approaches include: wave diffraction theory, the finite difference method, and the semi-analytical finite element method.[139]

Wave diffraction theory is a fundamental analytical method used to investigate wave interactions with vertical cylindrical structures. Eatock Taylor and Chau[140] developed an analytical wave diffraction model for submerged cylinders in finite water depths, based on the velocity potential function. Their approach computes hydrodynamic forces using variable separation techniques with eigenfunction expansions and validates computational accuracy through comparison with a higher-order boundary element method.

The finite difference method is particularly effective for modeling complex wave-structure interactions. Fu *et al.*[141] proposed a mesh-independent generalized finite difference method that incorporates an infinite-domain truncation technique, to simulate water wave interactions with multi-base cylindrical arrays. Their method adaptively selects the discrete node support set, thereby overcoming the mesh rigidity limitations of the traditional finite difference method and enabling its application to irregular geometries.

The semi-analytical finite element method offers significant advantages in analyzing structural elasticity and fluid-structure coupling effects. Seyfaddini *et al.*[142] developed a modeling framework that integrates the stochastic matrix model with a semi-analytical finite element method to examine the interaction between fluid flow and elastic cylinders composed of heterogeneous materials.

Traditional analytical methods, which are primarily based on linear wave theory, exhibit limited predictive accuracy under strong or extreme wave conditions. As wave amplitude increases, nonlinear hydrodynamic effects become increasingly significant, and linear models often fail to adequately capture the associated complex flow behaviors and force dynamics. To enhance the applicability of analytical approaches in realistic marine environments, future research should prioritize the refinement of analytical frameworks to better account for nonlinear wave-structure interactions.

3. CFD method

While analytical methods provide valuable theoretical insights, support structures in real-world engineering applications are often subjected to highly nonlinear flow environments influenced by a range of dynamic factors, including fluid-structure interaction, free-surface effects, and complex turbulence. Under such conditions, traditional analytical approaches frequently fall short of meeting the accuracy and flexibility requirements of engineering design and analysis. In contrast, CFD has emerged as a more precise and versatile tool for hydrodynamic analysis. By numerical solving the N–S equations, CFD allows for high-resolution simulations that capture the intricate flow phenomena surrounding tidal current energy device support structures.

In the development of the CFD method, the RANS equation has been widely used due to their favorable balance between computational efficiency and predictive accuracy. Diaz-Carrasco *et al.*[143] developed a numerical model for analyzing wave-structure-seabed interaction using the RANS framework. Their simulations successfully captured the force characteristics exerted by wave action on the structure. Similarly, Opoku *et al.*[144] applied the RANS equations to assess the hydrodynamic performance of an oscillating water column wave energy converter. Their study confirmed that the CFD simulations could accurately predict the device's hydrodynamic behavior, with results showing a high degree of agreement with experimental data.

CFD can be further enhanced through integration with theoretical and turbulence models, thereby improving the accuracy and robustness of hydrodynamic simulations. For example, Dutta *et al.*[145] combined CFD with an advanced turbulence modeling approach to perform three-dimensional numerical simulations of the hydrodynamic behavior of a square column subjected to turbulent flow. Beyond turbulence modeling, CFD can also resolve intricate aspects of fluid-structure interactions, particularly when multiple hydrodynamic effects act simultaneously. Aniszewski *et al.*[146] integrated CFD with the exact conservative level set method to simulate the interaction between a partially submerged solid structure and incoming waves. By employing an interface tracking technique, their model successfully captured the fine-scale details of fluid-solid coupling, including wave impact, free-surface deformation, and pressure distribution across the structure.

Given that numerical simulations often involve solving complex nonlinear equations and modeling multiphysics coupling phenomena, they typically demand substantial computational resources. To address these challenges, future research should focus on the integration of numerical optimization techniques with computationally efficient turbulence models. This approach aims to enhance computational efficiency while preserving simulation accuracy, thereby broadening the applicability of advanced numerical methods in practical engineering scenarios.

C. Coupled hydrodynamic analysis of the turbine-support structure system

In tidal current energy devices, the hydrodynamic interaction between the turbine and the support structure plays a critical role in determining operational efficiency, load distribution, and overall structural integrity. However, current research on turbine-support structure coupling has primarily focused on the unidirectional influence of the support structure on the turbine performance. The modeling and analysis of this coupled system vary significantly depending on the complexity of the marine environment. In simplified scenarios, assumptions such as uniform inflow, low turbulence intensity, or the neglect of wave effects are often employed to expedite analysis. In contrast, complex marine environments require consideration of non-uniform flow velocity profiles, wave-current coupling, and high turbulence intensity. These factors significantly complicate the interaction between the turbine and support structure, necessitating the use of high-fidelity numerical simulations or experimental investigations to achieve accurate and reliable results. Existing studies typically employ one or more of the following approaches for coupled analysis: Potential flow theory, CFD-based viscous flow analysis, and experimental methods. Each approach offers distinct advantages and is best suited to specific operating conditions, depending on the desired balance between computational cost, modeling complexity, and accuracy.[147–149]

1. Coupled analysis methods based on the potential flow theoretical framework

The potential flow theory approach addresses turbine-support structure interactions by applying momentum balance principles within an inviscid and irrotational flow field. This approach offers high computational efficiency and is particularly well-suited for preliminary assessments, such as evaluating the influence of the support structure on turbine thrust and power output. However, potential flow theory neglects viscous effects and is unable to accurately capture turbulence, wake interference, or vortex shedding phenomena. As a result, its applicability is significantly limited in complex marine environments, where high turbulence intensity and nonlinear flow structures dominate the turbine-structure interaction dynamics.

In a simplified oceanic environment, Satish[150] integrated the BEM method with the actuator disk method to analyze the dynamic response of a floating tidal current energy turbine. This study investigates the effects of surge and longitudinal motion of the floating support structure on turbine thrust and power output, under the assumption of a constant inflow velocity, while neglecting turbulence and wave-induced disturbances. The results indicate that the periodic motion of the support structure induces fluctuations in thrust loading, which in turn affect the stability and consistency of turbine power output.

In a complex marine environment, Pang *et al.*[151] investigated the effects of non-uniform inlet velocity profiles and turbulence intensity on the wake evolution of tidal current energy turbines using a coupled BEM and actuator disk model. The study found that vertical velocity gradients and localized vortex structures can cause uneven thrust distribution on the turbine. The results in pronounced asymmetric disturbances and delayed velocity recovery in the wake region. Additionally, the support structure acts as a localized flow disturbance, further intensifying these effects. It not only alters the loading conditions on upstream turbines but also increases power output fluctuations in downstream devices, ultimately reducing the overall energy capture efficiency of the turbine array.

2. Coupled analysis methods based on viscous flow CFD

CFD provides a more accurate representation of the fluid dynamics in complex environments compared to the traditional potential flow methods. With ongoing advancements in computational power and numerical algorithms. CFD is particularly well-suited for analyzing high Reynolds number flows, wake interference, and transient coupling effects, thereby enabling detailed assessments of system performance and structural loading in realistic oceanic conditions.[152]

In a simplified oceanic environment, Rehman *et al.*[153] conducted CFD simulations using the RANS method to systematically investigate the influence of different support tower diameters on the performance of HATTs and their wake structures. The results indicate that as the tower diameter increases, the shading effect becomes more pronounced. This leads to a significant rise in the amplitude of periodic fluctuations in the turbine's power output, highlighting the pronounced impact of tower-induced wake interference. In a related study, Marsh *et al.*[127] applied 3D unsteady RANS simulations to analyze the effects of various support structure configurations on torque pulsations in a VATT. Their findings demonstrated that the geometry of the support structure significantly affects local blade loading and plays a key role in determining the stability of the power coefficient.

In a complex marine environment, Wang *et al.*[154] systematically investigated the coupling effect of support structure on the performance of turbine and diffuser in diffuser-enhanced hydraulic turbine based on CFD method. The results show that the support structure significantly affects the local flow field acceleration capability and energy capture efficiency in a highly turbulent background. Badshah *et al.*[155] conducted a coupled fluid-structure interaction analysis to examine turbine-structure interactions under non-uniform flow velocity profiles. Their findings revealed that highly turbulent conditions significantly accelerate the accumulation of fatigue loads on the turbine blades, ultimately affecting the operational lifespan and structural integrity of the turbine system.

3. Coupled analysis based on experimental methods

Experimental methods provide measurement data of tidal current energy devices operating in real-world physical environments, serving as a valuable complement to numerical simulations. These methods help address the limitations of purely computational approaches, particularly in scenarios involving complex nonlinear hydrodynamic phenomena that are challenging to model accurately.

In a simplified marine environment, Mei *et al.*[156] investigated the effects of support structure pitching motion on turbine power output through circulating flume experiments. The experiments were conducted under constant flow conditions to evaluate the influence of various pitching periods and amplitudes on power output fluctuations. The results indicate that periodic variations in the angle of attack, caused by pitching motions, significantly affect the stability of power generation.

In a complex marine environment, Murray[157] conducted an experimental investigation into the bending-twisting coupling effect of flexible turbine blades. The results demonstrated that the dynamic deformation of flexible blades helps to alleviate the effects of non-uniform loading induced by the support structure, thereby enhancing the stability of power output.

A comprehensive coupled hydrodynamic analysis of the turbine-support structure system is essential for the optimal design and performance enhancement of tidal current energy devices. Each methodological approach offers distinct advantages: Potential flow theory is well-suited for rapid preliminary assessments, CFD provides high-resolution insights under complex sea state conditions, and experimental methods deliver critical empirical validation. To advance the reliability and efficiency of these systems, future research should focus on the integrated application of these three approaches to optimize tidal current energy device performance and enhance their operational stability and durability.

V. CHALLENGES IN OCEAN TIDAL CURRENT ENERGY TECHNOLOGY

A. Technological feasibility

Despite ongoing advancements, the practical deployment of ocean tidal current energy systems continues to face several technical challenges. Although the technology has reached a certain level of maturity, achieving large-scale implementation, effective array configuration, and intelligent operational control remains a critical challenge in its development. Ocean tidal current energy resources are abundant, and their efficient utilization depends on the deployment of large-scale individual devices or multi-unit arrays. However, large-scale systems involve high manufacturing and installation costs, while interactions between turbines within an array can result in overall power generation efficiency. While intelligent technologies offer the potential to improve environmental adaptability and operational performance, current limitations in real-time data acquisition, processing accuracy, and response speed pose significant barriers. These capabilities must be further enhanced to reliably support operations in highly dynamic and complex marine environments.[158–162]

B. Cost management

Despite the significant potential of ocean tidal current energy, high initial investment and O&M costs remain major barriers to its widespread adoption.[115] The considerable expenses associated with equipment manufacturing, offshore installation, and long-term maintenance directly affect the economic feasibility of tidal current energy projects. In addition, substantial ongoing investment is required to support routine maintenance and system upgrades throughout the operational lifespan of the devices. Therefore, the development of effective cost control strategies is essential to improving the lifecycle economics of these systems. Reducing capital and operational costs remains a critical priority for the continued advancement and commercial competitiveness of tidal current energy technologies.

C. Environmental impact

As a widely recognized renewable energy, ocean tidal current energy holds significant potential for sustainable development. However, its ecological impacts remain insufficiently understood and warrant comprehensive assessment to ensure alignment with environmental protection goals. Throughout the lifecycle of tidal current energy devices, various stages may pose ecological risks. For example, installation activities can alter marine habitats and disturb local current patterns, while operational noise may disturb the behavior and communication of marine organisms. Furthermore, noise emissions and the presence of rotating components during operation may interfere with the behavior and spatial distribution of marine organisms. To more accurately evaluate these environmental risks, case-specific studies based on field data have been conducted. For instance, Copping *et al.*[163] reported that approximately 20% of Clupea pallasii individuals in the Bay of Fundy, Canada, may enter the potential impact zone of tidal turbines during specific tidal stages. Similarly, Peraza and Horne[164] developed a conditional probabilistic model using empirical data from Admiralty Inlet, demonstrating that under nighttime conditions with no avoidance behavior, the probability of fish encountering and being impacted by tidal turbines could reach as high as 0.689.

Furthermore, the presence of tidal current energy devices can modify flow fields, turbulence structures, residence times, and seabed shear stress, potentially affecting coastal hydrodynamics and related biogeochemical processes—including sediment transport, nutrient cycling, and water quality dynamics.[165] Currently, these environmental effects remain understudied, and a lack of long-term field data limits our ability to predict and manage them effectively. As such, balancing the deployment of tidal current energy technologies with the need to preserve marine ecosystems represents a critical challenge.

VI. TRENDS AND CHALLENGES IN OCEAN TIDAL CURRENT ENERGY TECHNOLOGY

A. Array deployment

(1) **Trend:** A well-optimized array layout plays a crucial role in maximizing tidal current resource utilization and improving the overall power generation capacity. Within an array, tidal current energy devices can share infrastructure—including support structures and submarine cables—which facilitates cost reduction through economies of scale, mass production, and streamlined installation procedures.[166–169] In practice, the deployment of a single 100 MW tidal current energy system poses significantly greater technical and logistical complexity than the installation of 100 individual 1 MW units. Array-based deployment is not only more cost-effective but also enhances operational reliability. In the event of a unit failure, the remaining turbines in the array can continue functioning independently, whereas a centralized large-scale system may require a complete operational shutdown during maintenance or repair.[170–172]
The design of tidal current energy arrays involves both macro-scale and micro-scale considerations. At the macro-scale, design decisions focus on determining the total number of turbines and their overall spatial arrangement within the deployment site. At the micro-scale, efforts are directed toward optimizing the relative positioning and inter-turbine spacing to minimize wake interference and maximize energy capture efficiency. To support the advancement of array deployment strategies, researchers are actively exploring innovative configuration schemes and evaluating their performance through numerical modeling and simulation.[173] In addition to layout optimization, ongoing research is increasingly focusing on yaw angle control, controller design, and the detailed analysis of wake characteristics analysis.[174]

(2) **Challenge:** In regions characterized by significant tidal current variability and complex sea states, accurately assessing the interactions among arrayed devices remains a major challenge. These hydrodynamic interactions, including wake effects, turbulence propagation, and flow field distortion, introduce uncertainties in power generation efficiency predictions, complicating both design and performance optimization. Moreover, tidal current energy arrays require extensive submarine infrastructure, including power transmission systems, cabling networks, and mooring components. The construction, operation, and maintenance of such systems demand rigorous environmental monitoring and adherence to strict regulatory frameworks, particularly in ecologically sensitive marine zones. These requirements inevitably contribute to increased system complexity and cost, posing a significant barrier to the scalability and economic viability of large-scale tidal current energy deployments.[175,176]

B. Intelligent operation

(1) **Trend:** In recent years, artificial intelligence algorithms have been increasingly applied to support site selection, tidal current energy forecasting, and the dynamic control of tidal turbines. Under complex sea states and in deep-sea environments, it

remains difficult for conventional systems to optimize power generation performance. In contrast, intelligent systems can continuously monitor both the marine environment and equipment status in real time, allowing for autonomous adjustments to operating modes in response to tidal current fluctuations, thereby enhancing overall energy output.[177] Furthermore, intelligent systems equipped with machine learning capabilities can analyze historical performance data to predict potential equipment failures and facilitate proactive maintenance planning. This predictive approach reduces unplanned downtime, improves operational efficiency, and minimizes the need for human intervention, contributing to lower maintenance costs and increased system reliability.[178]

(2) **Challenge:** The sensors and control systems integrated into intelligent tidal current energy devices must demonstrate long-term operational stability under extreme marine conditions. These systems are particularly vulnerable to tidal fluctuations, salt spray corrosion, and biofouling, all of which can compromise their performance over time. Maintaining accurate data acquisition and ensuring reliable real-time transmission in such complex and dynamic environments presents a fundamental challenge in system design. Furthermore, the high cost and technical complexity associated with intelligent systems significantly increase the initial investment requirements and contribute to elevated operational and financial risks, potentially hindering large-scale deployment and commercial viability.[179,180]

C. Economic

(1) **Trend:** The adoption of technologies such as array deployment, modular design, and artificial intelligence algorithms has progressively contributed to the reduction of maintenance costs for tidal current energy devices. At the same time, improvements in design optimization, the use of advanced materials, and economies of scale have helped to significantly lower installation costs. These cost reductions are expected to enhance the market competitiveness of tidal current energy, support large-scale deployment, and accelerate the transition toward green energy systems. In addition to addressing capital and operational cost reductions, Matteo *et al.* quantified the energy potential of tidal current energy devices and developed an economic evaluation model incorporating LCOE, net present value, and discounted payback period. This model enables the estimation of costs across each stage of the device lifecycle and proposes targeted strategies to improve overall economic viability.[181]

(2) **Challenge:** Balancing cost reduction with the need to maintain device quality and long-term operational stability remains a significant challenge in the development of tidal current energy technology. The use of low-cost materials may compromise the durability and structural integrity of devices, especially in high-salinity, corrosion-prone marine environments. Moreover, the high initial capital investment required for tidal current energy installations—particularly for emerging technologies such as array deployment—poses difficulties in reconciling short-term financial constraints with long-term economic gains. At present, comprehensive economic models integrating multiple factors remain underdeveloped.

D. Sustainable development

(1) **Trend:** Sustainable development is not only a core objective of tidal current energy technology but also a key priority in modern societal progress. To mitigate the impacts, researchers have focused on understanding tidal currents dynamics and turbine-generated operation noise, while optimizing device siting and wake characteristics to reduce disturbances to marine ecosystems.[182] As the scale of tidal current energy arrays continues to grow, their compact configurations may amplify cumulative environmental effects, making array layout optimization an increasingly important area of research.[183] In addition, achieving true lifecycle sustainability requires improvements in the manufacturing process, including the use of environmentally friendly materials, enhancements in energy efficiency, and strategies to maximize resource recovery and recyclability.

(2) **Challenge:** Balancing energy production with environmental conservation remains a major challenge in the advancement of sustainable tidal current energy development. Ongoing research efforts aim to reduce acoustic emissions, vibrational impacts, and disruptions to marine habitats through strategies such as hydrodynamic blade shape optimization, the use of environmentally friendly materials, and improvements in array layout and site selection. However, many of these measures remain largely theoretical and have not yet been widely implemented in practice. As tidal current energy emerges as a promising renewable resource, there is an urgent need for comprehensive, long-term ecological assessments to better understand its impacts on marine ecosystems and to develop effective strategies for mitigation and environmental integration.

VII. PERSPECTIVES ON OCEAN TIDAL CURRENT ENERGY TECHNOLOGY

A. Technological maturity

With continuous advancements in research and successive technological iterations, ocean tidal current energy technology is rapidly progressing toward engineering-scale implementation and intelligent operation. Notable progress has been made in areas such as hydrodynamic model optimization, the integration of intelligent control systems, and the structural design of tidal current energy devices. To overcome current technological bottlenecks, future efforts should focus on refining high-precision fluid–structure interaction models, developing corrosion-resistant composite materials, and enhancing the environmental adaptability of intelligent control systems. These improvements are essential for ensuring stable, reliable, and efficient operation in complex and variable marine environments. In support of this transition, many manufacturers have already begun implementing digital twin technology in their production and monitoring processes, enabling real-time diagnostic, predictive maintenance, and performance optimization through virtual system replication.

In the future, tidal current energy technology can be further extended to support operation and maintenance through the integration of virtual-reality interaction systems. Such technologies will enable real-time power generation control, fault prediction,[184] and autonomous system optimization, thereby improving energy conversion efficiency and reducing operation and maintenance costs. Moreover, continued technological innovation will drive industry chain integration and accelerate the commercialization and scaling of the sector. The adoption of modular design principles, the establishment of robust international standardization frameworks, and the accumulation of large-scale deployment experience will be essential in mitigating technical risks. Collectively, these advancements will enhance the market competitiveness of tidal current energy and position it as a strategic pillar within the broader renewable energy sector.

B. Cost controllable

Advancements in manufacturing processes, the adoption of advanced materials, and the optimization of supply chains have collectively contributed to the ongoing reduction of manufacturing, installation, operation, and maintenance costs associated with tidal current energy devices. At the same time, large-scale production and design optimization have significantly improved the energy conversion efficiency of these systems, providing strong technical support for both cost reduction and high-yield energy output.

In the future, continued advancements in manufacturing technologies, the adoption of next-generation materials, and further improvements in supply chain management are expected to drive additional cost reductions, enhancing the economic viability of tidal current energy systems and improving their competitiveness relative to other renewable energy sources. Moreover, economic models based on lifecycle cost analysis are currently being developed to support more refined and data-driven cost control strategies. When integrated with intelligent technologies, these systems will further improve economic efficiency, thereby accelerating the commercial-scale deployment of tidal current solutions.

C. Environmental and social benefits

As an essential component of the green and clean energy portfolio, ocean tidal current energy is poised to play an increasingly important role in promoting both environmental sustainability and socioeconomic development. Looking ahead, the design and operation of tidal current energy devices will place greater emphasis on environmental stewardship, with the goal of minimizing their impact on marine ecosystems. To this end, strategies such as optimized installation layouts, the use of eco-friendly materials, and improvements in energy conversion efficiency will be key to reducing habitat disruption and mitigating the ecological effects of operational noise, vibration, and turbulence.

The widespread adoption of ocean tidal current energy will contribute significantly to climate change mitigation, the reduction of greenhouse gas emissions, and the advancement of a more diversified and resilient global energy mix. Tidal current energy devices offer a stable and reliable power supply, particularly well-suited for coastal regions, while also promoting regional economic development, generating employment opportunities, and enhancing overall social well-being.[182] Moreover, the sustainable development of ocean tidal current energy aligns closely with the United Nations sustainable development goals and represents a critical enabler of the global shift toward a green, low-carbon economy. By integrating environmental, economic, and social dimensions, tidal current energy technology has the potential to serve as a driving force for a more sustainable and equitable energy future.[185–188]

VIII. CONCLUSIONS

This paper provides a comprehensive review of the current state of tidal current energy technology, covering its applications, hydrodynamic analysis methods, key technical challenges, development trends, and the broader challenges associated with its implementation. Additionally, it offers insights into the future trajectory of the field. With ongoing technological advancements, the development of tidal current energy is gradually transitioning from the experimental validation stage to commercial-scale deployment. Nevertheless, its progress is still limited by key factors, including efficiency optimization, cost reduction, environmental considerations, and the need for stronger policy and regulatory support. To enable the large-scale deployment and sustainable development of tidal current energy systems, future research must adopt an integrated approach that addresses both technical and non-technical dimensions. This includes advancements in engineering design, materials science, intelligent control, as well as economic modeling, environmental impact assessment, and regulatory frameworks. The key conclusions of this study are as follows:

(1) **Types of tidal current energy devices:** Tidal current energy technologies have evolved from early experimental prototypes to commercial applications and large-scale deployments. At present, the primary types of tidal current energy devices include HATTs, VATTs, oscillating hydrofoils, and ducted turbines, each offering distinct technical features and application scenarios. HATTs are characterized by high energy conversion efficiency and operational stability, but their deployment involves complex installation procedures and intensive maintenance requirements. VATTs exhibit strong bidirectional flow adaptability and simplified structural integration but typically suffer from lower efficiency and more complex loading conditions. Oscillating hydrofoils are valued for their mechanical simplicity and low manufacturing cost, through their limited power output and restricted site suitability constrain broader application. Ducted turbines improve efficiency by accelerating local flow velocity through a Venturi-like structure, but they perform optimally only under stable, highly directional flow conditions.

(2) **Hydrodynamic analysis methods:** Hydrodynamic analysis methods for tidal current energy devices have undergone continuous refinement, encompassing turbine performance evaluation, support structure analysis, and turbine-structure coupling assessments. For turbine analysis, CFD provides high computational accuracy and effectively captures nonlinear flow characteristics. However, due to its high computational cost, CFD is best suited for small-scale optimization and high-fidelity design studies. In contrast, methods such as BEM, vortex method, and ALM are commonly applied in early-stage design and parametric analysis, though they exhibit limited accuracy in complex or transient flow conditions. For support structure analysis, the Morison equation remains widely used due to its computational simplicity, particularly in low Reynolds number regimes. However, it becomes less reliable under high Reynolds number flow or strong wave-current interaction. Analytical methods, such as wave diffraction theory and semi-analytical models, provide theoretical accuracy in idealized conditions but struggle to capture nonlinear hydrodynamics. CFD-based approaches

are increasingly adopted for support structures, as they allow for detailed modeling of turbulence, vortex shedding, and wave-structure interactions in realistic marine environments. For turbine-support structure coupling analysis, potential flow theory is primarily used in simplified settings, providing rapid estimations of interaction effects. In contrast, CFD coupled with fluid-structure interaction techniques offers more comprehensive and accurate assessments in complex marine conditions, accounting for unsteady loads, dynamic motion, and structural responses under realistic sea states.

(3) **Key technical issues:** Tidal current energy technology has seen significant advancements in recent years. However, its large-scale deployment remains constrained by several critical challenges, particularly in the area of technical feasibility, cost-effectiveness, and environmental sustainability. From a technical perspective, while tidal current energy devices have reached a relatively advanced stage of development. Their operation in complex marine environments still presents significant difficulties. Key issues include the optimization of array configurations and the implementation of intelligent operational systems. Challenges such as high equipment manufacturing costs, efficiency losses due to inter-device interference within arrays, and the limited responsiveness of real-time control systems—especially in terms of data acquisition and processing speed—hinder effective deployment. On the economic front, the high initial investment and considerable lifecycle operation and maintenance costs remain major barriers to the commercialization of tidal current energy technologies. There is a pressing need for integrated cost-control strategies that span all phases of system deployment—including installation, operation, and maintenance—to enhance the overall economic viability of these systems. In terms of environmental impact, tidal current energy devices have potential to affect marine ecosystems through mechanisms such as habitat alteration, disruptions of natural current patterns, and modification of sediment transport dynamics. Given the current limited scope of research in this area, it is crucial to strengthen environmental impact assessments and develop targeted mitigation strategies. These efforts must proceed in parallel with technological innovation to ensure alignment with long-term sustainability goals.

(4) **Development trends and challenges:** The future development of tidal current energy technology will focus on advancing intelligent operation, optimizing array deployment, improving economic feasibility, and promoting environmentally sustainable design. The integration of artificial intelligence and smart technologies will enable real-time monitoring, adaptive control, and predictive maintenance, thereby enhancing power generation efficiency and reducing operation and maintenance costs. In parallel, array deployment strategies will focus on optimizing turbine arrangement, mitigating wake interference, and maximizing overall energy output. However, significant challenges remain in managing hydrodynamic interactions among turbines and in ensuring efficient and coordinated power transmission across large-scale arrays. Achieving economic feasibility will require continued efforts to reduce the costs of equipment manufacturing, installation, and lifecycle operation and maintenance, while improving energy conversion efficiency. These improvements are essential for enabling market-driven, commercial-scale deployment. From an environmental perspective, future tidal current energy systems must be designed to minimize ecological impacts, including disruptions to marine habitats, flow dynamics, and benthic ecosystems. As such, comprehensive environmental impact assessments and the development of ecological compensation mechanisms will be critical areas of research to ensure the sustainable integration of tidal current energy into the broader marine environment.

Finally, as an emerging renewable energy source, tidal current energy holds substantial potential to support the transition toward a low-carbon energy structure. However, its industrial-scale development remains highly dependent on policy support, due to high initial investment and the ongoing demand for technological innovation. Policy frameworks play a crucial role in advancing the tidal current energy sector. Governments can help reduce investment risks and stimulate technological progress by implementing targeted measures such as tax incentives, financial subsidies, and increased funding for research and development. Furthermore, the establishment of renewable energy tariff subsidies and participation in carbon trading markets can enhance the economic competitiveness of the tidal current energy within the broader renewable energy landscape. At the same time, it is essential for governments to establish stringent environmental standards and robust regulatory frameworks to prevent potential irreversible impacts on marine ecosystems associated with the development of tidal current energy.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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